Information Processing and Management 59 (2022) 102909



Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm





Parkinson's disease diagnosis using neural networks: Survey and comprehensive evaluation

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ARTICLE INFO

Keywords:
Parkinson's disease
Deep learning
Neural networks
Multi-modal learning
Machine Learning
Diagnosis

ABSTRACT

Parkinson's disease (PD) is a chronic neurodegenerative disease of that predominantly affects the elderly in today's world. For the diagnosis of the early stages of PD, effective and powerful automated techniques are needed by recent enabling technologies as a tool. In this study, we present a comprehensive review of papers from 2013 to 2021 on the diagnosis of PD and its subtypes using artificial neural networks (ANNs) and deep neural networks (DNNs). We present detailed information and analysis regarding the usage of various modalities, datasets, architectures and experimental configurations in a succinct manner. We also present an in-depth comparative analysis of various proposed architectures. Finally, we present a number of relevant future directions for researchers in this area.

1. Introduction

Parkinson's disease (PD) is one of the most chronic neurodegenerative diseases in today's world as it affects 2–3% of the 65-year-old population (Poewe et al., 2017; Radhakrishnan & Goyal, 2018). PD is a prototypical movement disorder, and James Parkinson first described it in 1817 (Menken, Janca, Organization, et al., 1997). The primary symptoms of PD are: (*i*) tremor, (*ii*) rigidity or muscle stiffness, (*iii*) bradykinesia and (*iv*) postural instability (*Parkinsons-Disease-Information-Page, (Accessed on : 15 June 2020)*, 2020) and these symptoms are generally known as Parkinsonism Syndrome. In recent years, there has been a significant increase in the use of machine learning based computer-aided diagnosis (CAD) systems to diagnose diseases, sometimes even in early stages (Gupta, Kumar, Arora, & Raman, 2021; Kumar, Gupta, Arora, & Raman, 2021a, 2021b; Masud et al., 2021). There has also been an increase in utilization of such CAD systems for diagnosing PD from various modalities like speech signals (Hariharan, Polat, & Sindhu, 2014; Zia, Tadayon, McDaniel, & Panchanathan, 2016), gait signals (Del Din, Godfrey, & Rochester, 2015; Khorasani & Daliri, 2014), magnetic resonance imaging (MRI) (Bouza, Yang, & Vemuri, 2020; Esmaeilzadeh, Yang, & Adeli, 2018; Huang et al., 2021), positron emission tomography (PET) (Hwang et al., 2018), single-photon emission computed tomography (SPECT) (Grover, Bhartia, Yadav, & Seeja, 2018; Pianpanit et al., 2020), Dopamine Transporter Scan (DaT Scan) (Eskofier et al., 2016; Sivanesan, Anwar, Talwar, Menaka, & Karthik, 2016), tremor signal (Hosseini & Makki, 2013), handwriting signal, handwritten images (Pereira, Pereira, Papa, Rosa, & Yang, 2016) and various other clinical features (CF) (Er, Cetin, Bascil, & Temurtas, 2016).

Artificial neural networks (ANNs) and deep neural networks (DNNs) are special types of machine learning models that have found an increased adaptation for various CAD of many diseases. Such CAD systems have also been used to classify and monitor

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https://doi.org/10.1016/j.ipm.2022.102909
Received 16 August 2021; Received in revised form 1 February 2022; Accepted 14 February 2022
Available online 21 March 2022
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PD. Er et al. (2016) and Challa, Pagolu, Panda, and Majhi (2016) utilized fuzzy neural systems for prediction of PD. Naseer et al. (2020) and Ortiz et al. (2019) utilized convolutional neural networks (CNNs) in detection of PD, whereas recurrent neural network (RNNs) were used by Ashour, El-Attar, Dey, El-Kader, and El-Naby (2020) and Ribeiro, Afonso, and Papa (2019) for detection of freezing of gait (FoG) and diagnosis of PD respectively.

However, in the area of PD diagnosis using ANNs and DNNs, the information regarding various proposed methods and trends in the recent years is not available in a single place. This is due to lack of sources that can disseminate clear comparative analysis of proposed models, their performances, and experimental configurations. There are a few works (Khachnaoui, Mabrouk, & Khlifa, 2020; Watts, Khojandi, Shylo, & Ramdhani, 2020) that review some of the related techniques for machine learning (ML) and deep learning (DL) for CAD of PD. In order to bridge this gap, we present this research to provide readers with a single source of information that can convey about the recent trends in CAD of PD using ANNs and DNNs, meaningful insights into technological innovations, and important possible future directions to further progress in this area.

We designed this paper in a way that minimizes separate reading of the papers that have been covered in this paper. That is, a reader of this paper will find most of the relevant information in this paper and will not require to consult the original papers over and over again. We disseminate the information in the form of three tables that place most of the content of the papers reviewed here in a compressed and objective manner.

Table 1 gives a very high-level information about the models, datasets, number of subjects and cross validation techniques being used by the researchers. Table 2 contains information that goes much deeper than that contained in Table 1. The information contained in Table 2 contains the detailed network architecture and several other experimental configurations. A significant advantage of this design is that the reader can find most of the relevant information in a single place in the form of this review paper.

The main contributions of this paper are as follows:

1. We provide most relevant information collected from 143 papers published from 2013–2021 on diagnosis and classification of PD using ANNs and DNNs in a highly compact manner within this paper.
2. We disseminate meta-level analysis of the trends in the field that depicts the nature of modalities and datasets being used, the performance of every proposed architecture, and details regarding other experimental configurations followed in each paper.
3. We also perform analysis of individual technological details like the proposed ANN or DNN architecture, learning rate, hyperparameter tuning, normalization and regularization schemes used by the researchers.
4. We also provide detailed information regarding the pros and cons of the models proposed by researchers.
5. We also provide numerous future directions to help researchers in identifying several possible directions for them to work upon.

Loh, Hong, et al. (2021) also do a review of application of DL techniques for CAD of PD. this work is different from it due to the addition tables for describing the architectures and their pros and cons. The effect of these tables are to help the reader to find most of the relevant details along with ideas for future work in a single place.

The rest of the paper is designed as follows: Section 2 contains the search strategy used for finding relevant papers, Section 3 contains an entire survey of the literature whereas, Section 4 contains the discussion regarding papers covered in the literature survey. Section 5 contains future directions that can be followed, and finally, we present the conclusion in Section 6.

2. Search strategy

We searched prominent papers in the field from Google Scholar (https://scholar.google.co.in/) and ScienceDirect (https://www.sciencedirect.com/). This resulted in papers using a combination of ANN, RNN, CNN, Transfer learning, etc. We also included papers based on deep learning and ensemble methods for PD diagnosis. In this article, we have reviewed ANNs, CNNs, and RNNs from the period 2013–2021.

3. Literature survey

In this study, we have extensively reviewed the conventional neural network and deep learning techniques for PD diagnosis within three major categories based on the architecture of the model: (*i*) ANNs based architectures, (*ii*) CNNs based architectures and (*iii*) RNNs based architectures and their ensembles. The general technical information about each paper studied is mentioned in Table 1. More technical details about the papers covered in Table 1 are covered in Table 2 which provides detailed information about the proposed model's configuration.

3.1. ANN based architectures

In this section, we reviewed the work that covers the PD diagnosis using ANNs and its variants. Gharehchopogh and Mohammadi (2013) introduced two types of ANNs for the classification and diagnosis of PD using multi-layer perceptron (MLP) with back-propagation learning algorithm and Radial Basis Function (RBF), ANNs used to differentiate them based over UCI dataset and achieved an accuracy of 93.22% with MLP, and 86.44% with RBF-ANN. Hossen (2013) presented an ANN-based architecture for PD diagnosis. This work's primary purpose is to use an ANNs to select the best features and discriminate between the two types

of tremors using spectral analysis of tremor time-series recorded by accelerometer and surface electromyography (EMG) signals. Babu and Suresh (2013) presented a prediction system to predict the PD using gene expression dataset. Mandal and Sairam (2013) developed a statistical approach to enhance early-stage PD diagnosis using ML classifiers like sparse multinomial logistic regression (LR), support vector machine (SVM), ANN, Boosting algorithms and their ensembles. Hosseini and Makki (2013) proposed two modular techniques for PD detection and diagnosis. First, features were extracted and selected using local linear and non-linear principal component analysis (PCA). Finally, an ANN was used to classify the PD over the chosen features and reported an accuracy of 92% and 97.3% with first and second model, respectively. Similarly, Hariharan et al. (2014) proposed an intelligent hybrid system to detect the PD with healthy controls (HC). In this work, a three-step methodology was proposed: in the first step, the Gaussian mixture model (GMM) was used to preprocess the features, then feature reduction or selection techniques, namely principal component analysis (PCA), linear discriminant analysis (LDA), sequential forward feature selection (SFS) and selection backward feature selection (SBF) were used. Finally, ML supervised predictive model least-square support vector machine (LS-SVM), probabilistic neural networks (PNN), general regression neural network (GRNN) to detect the PD with an accuracy of 100%. Khorasani and Daliri (2014) proposed a hidden Markov model (HMM) based PD detection technique against healthy subject using gait data and reported an accurately classifying with more than 90% with a data of 31 patients.

Sharma and Giri (2014) presented an automatic discrimination technique of PD diagnosis against HCs using ANN and SVM predictive classifier over UCI-ML dataset (23 PD vs 8 HC) and achieved an accuracy of 85.294%. Beheshti et al. presented centripetal accelerated PSO and an improved version of ANN to recognize the different kind of disease, including PD and reported a mean accuracy of 92.56%, 92.56%, 82.05%, and 92.31% using centripetal accelerated particle swarm optimization (CAPSO)-MLP, particle swarm optimization (PSO)-MLP, Gravitational search algorithm (GSA)-MLP, and Imperialist competitive algorithm (ICA)-MLP, respectively. Islam, Parvez, Deng, and Goswami (2014) presented a comparative study for early detection and diagnosis of PD over voice disorder speech dataset and with various ML predictive classifiers such as Random Tree (RT), SVM and forward–backward artificial neural network (FB-ANN), trained the classifiers using cross-validation fashion and reported an accuracy of 97.37%. Olanrewaju, Sahari, Musa, and Hakiem (2014) proposed an early PD detection and diagnosis techniques using ANN as ML predictive classifiers and reported compression metrics like accuracy (ACC), sensitivity (SEN), specificity (SPEC) as 80%, 83.3%, 63.6%, respectively. Joo et al. (2014) suggested a predictive model by adopting ANN to develop the model and predict the gait velocity across the entire cycle in stance and swing phase using plantar pressure data. They were able to clarify the causes of PD, using a motion capture system by a five-fold cross-validation scheme, and were reported the gait correlation coefficients (CC) ⟨normal walking, slow walking, fast walking⟩ ≡ ⟨96.3%, 97.8%, 95%⟩, respectively. Shaikh (2014) presented PD and primary tumor prediction using predictive models and reported accuracy to detect PD with ML predictive classifiers ⟨ANN, DT, NB⟩ ≡ ⟨90.76%, 80.51%, 69.23%⟩, respectively. Mandal and Sairam (2014) proposed a new ML predictive algorithms for PD using ensembling of different ML classifiers and such as ANN, DT and using linear LR and sparse multinomial LR, reported performance metric ⟨Acc, Sens, Spec ⟩ ≡ ⟨100%, 98.3%, 99.6%⟩, respectively. Froelich, Wróbel, and Porwik (2014) investigated the PD diagnosis system, which was dependent on the signature characteristics of a person's speech, and performed compression for the classification task using various ML predictive model including ANN and reported an accuracy of 81.25% to classify the PD with ANN.

Davidson, de Paor, Cagnan, and Lowery (2015) presented PD classification techniques over brain stimulation data using a series of network models. Similarly, Peker, Sen, and Delen (2015) presented a novel approach for diagnosing PD using speech signal. First, features are extracted from the speech signal, a continuous phonation recordings, and minimum redundancy maximum relevance (mRMR) were applied to select relevant features, and finally, complex-valued artificial neural network (CVANN) were used for effective diagnosis of PD. Hirschauer, Adeli, and Buford (2015) presented a computer-assisted PD diagnosis system using an improved version of the probabilistic neural network (PNN) to classify the PD vs HC. In this article, author also presented a multi-class classification using probabilistic neural network to classify PD, HC, and scans without evidence for dopaminergic deficit (SWEDD) patients over the Parkinson's Progression Markers Initiative (PPMI) database. The author has reported an accuracy of 98.6% to classify HC versus PD Patient. Saloni, Sharma, and Gupta (2015) presented a telediagnosis of PD with an analysis of voice signal using ML predictive classifiers such as SVM and ANN achieved an accuracy of 95.9% and 87.7% with SVM and ANN, respectively.

Jane, Nehemiah, and Arputharaj (2016) presented a temporal classification model namely Q-backpropagated time-delay neural network (Q-BTDNN) for PD diagnosis by using the gait disturbances dataset and reported an accuracy of 91.49%, 92.19% and 90.91%. Khan, Chalup, and Mendes (2016) presented a new neuro-evolutionary algorithm based on wavelet neural network (WNN) by using Cartesian genetic programming to classify the PD over the speech signal dataset and reported a promising result. Al-Fatlawi, Jabardi, and Ling (2016) has adopted a deep belief network to diagnosis the PD by distinguishing and analyzing speech signal and achieved an accuracy of 94% to differentiate between PD patients and healthy people. Pereira et al. (2016) proposed a methodology to classify the PD patients by deploying the CNNs model for extracting features from handwritten (HandPD) exams. Authors have implemented some fine-tuning with hyperparameters using BAT algorithm, firefly algorithm and particle swarm optimization (PSO) technique and reported an accuracy of 97.42% to classify the PD. In this work (Turner et al., 2016), the author implemented a new regulatory artificial gene network named as epiNet and used a time series data monitoring collected from patients who had gone through levodopa therapy for PD. Abiyev and Abizade (2016) presented a hybrid technique that combines the fuzzy system and neural networks, named as fuzzy neural systems (FNS) to distinguish the people with PD from HC subjects and reported an accuracy of 100%. Benba, Jilbab, and Hammouch (2016) presented PD classification techniques by using the analysis of speech signals recorded from 40 subjects (20 PD vs 20 Normal subjects). First of all, authors have extracted the voice-print from each of the signals, and by using the techniques, namely Mel Frequency Cepstral Coefficients (MFCCs) and then compressed it by averaging them. Finally, SVM was used to classify it and reported an accuracy of 82.5% over the dataset as mentioned above, and 100% over an independent dataset of that is consisted of 28 PD patients. Sivanesan et al. (2016) presented a decision-support system to assist the

doctor in deciding whether the patient is healthy or under risk of PD. The proposed work combines eye-hand coordination behaviour with the DaT Scan images in order to determine the risk of this disorder. Initially, the patient's eye-hand coordination level is assessed through a hardware module; then, the DaT Scan image is analysed and used to extract specific geometrical parameters that indicate the presence of PD. These parameters are finally fed into a Multi-Layer Perceptron Neural Network using Levenberg–Marquardt (LM) backpropagation training algorithm. Experimental results indicate that the proposed system exhibits an accuracy of 93%. Zia et al. (2016) used a neural network approach to detect PD symptoms, named as freezing of gait (FoG) by using the public available UCI FoG experimental dataset.

Challa et al. (2016) have presented model using machine learning techniques ⟨ MLP, Bayes network (BayesNet), random forest (RF) and Boosted LR ⟩ for early prediction of PD to help the doctor to take the necessary pre-treatment action and reported an accuracy of 97.15% with Boosted LR and receiver operating characteristic (ROC) of 98.9%. Er et al. (2016) conducted a comparative analysis for PD using different types of ML classifier, and compared the effectiveness of different types of classification techniques to effectively detect the PD. Hlavica, Prauzek, Peterek, and Musilek (2016) contributed a PD assessment methodology using computational methods. First of all, the author has extracted the CF from the speech signal, then using the computational model, namely ANNs and adaptive network-based fuzzy inference system (ANFIS) for classifying whether a patient is suffering from PD or not. Guruler (Gürüler, 2017) proposed a novel complex-valued ANN (CVANN) for PD diagnosis, which employed a k-means clustering feature weighting method.

Gómez-Vilda et al. (2017) extracted features from speech articulation movements and passed the extracted features to an ANN for PD detection. Nikandish and Kheirkhah (2017) used a two-layer MLP for PD classification using features extracted from gait signals. Rumman, Tasneem, Farzana, Pavel, and Alam (2018) processed features extracted from the Caudate and Putamen regions for PD classification using ANN. Parisi, RaviChandran, and Manaog (2018) used ANN for feature selection and SVM for classification from features extracted on speech data. Khan, Mendes, and Chalup (2018) implemented an ensemble of evolutionary wavelet neural network (EWNN) for breast cancer and PD prediction. Wroge et al. (2018) utilized an ANN for PD diagnosis using features extracted from voice data. Grover et al. (2018) proposed a three layer deep MLP for predicting severe and non-severe PD patients from voice data based in total UPDRS and UPDRS scores.

Johri and Tripathi (2019) used CNN on image spectrograms generated from the gait data and used a multilayer perceptron (MLP) for the diagnosis of Parkinson patients. A class weight measure was also used for the MLP to resolve the PD's class imbalance problem. Mostafa et al. (2019) proposed a multiple feature evaluation algorithm (MFEA) to rank relevant features. These features are evaluated with multiple classification methods, in which ANN outperformed other classifiers. Ali, Zhu, Zhang, and Liu (2019) implemented a genetically optimized ANN on features extracted from voice data for automated classification of PD. Mikos et al. (2019) used ANN for real time PD classification on gait signals from a wearable device.

Vital, Murty, Rao, and Sriram (2020) utilized statistical features and ANN to classify PD. The ANN was consisting of 10 hidden units and achieved a classification accuracy of 100%. Liu et al. (2020) has designed a backpropagation neural network (BPNN) controller based Closed-Loop Deep Brain Stimulation to readjust the proportional–integral–derivative (PID) parameters for different PD states and approximately 58.26% energy consumption saved through the controller. Rasheed et al. (2020) presented a PD prediction using variable adaptive moment-based backpropagation algorithm of ANN named as BPVAM over Oxford PD detection dataset (Little, McSharry, Roberts, Costello, & Moroz, 2007) and attained an accuracy of 97.50%. Wanjale, Nagapurkar, Kaldate, Kumbhar, and Bala (2020) introduced a Severity prediction of PD using ANN and reported an accuracy of 83.56% with a hold-out scheme. Chakraborty, Aich, and Kim (2020) presented PD detection techniques using textural and statistical features extracted from MRI scans and classify it using ANNs, and reported an accuracy, recall, precision and f1-score are 95.3%, 95.41%, 97.28% and 94%, respectively.

Veeraragavan, Gopalai, Gouwanda, and Ahmad (2020) has introduced Vertical Ground Reaction Force (VGRF) to extract and select the relevant features from PD GAIT dataset and input to the ANNs and achieved a high accuracy of 97.4%. In Olivares et al. (2020a), authors proposed extreme learning machine (ELM) by using the Bat Algorithm, to detect the PD and reported an accuracy of 96.74% with PD audio dataset taken from UCI Machine Learning Repository. Huang et al. (2020) proposed a multi-stage prediction of PD using 196 subjects with SPECT imaging. First, they used several feature extraction techniques such as PCA, multilinear PCA, intensity summary statistics (IStat), and Laws' texture energy measure to extract the features from Brain image dataset then extracted features are input to the several machine and deep learning predictive model such as RF, AdaBoost and Deep CNN, to predict the stage of the PD disease and reported an accuracy of 64.9% with Deep CNNs.

Chien, Hsu, Lee, Sung, and Lin (2021a) used abnormal DAT-SPECT images of subjects with PD and parkinsonism caused by other disorders were divided into training and test sets. Striatal regions of the images were segmented by using an active contour model and were used as the data to perform transfer learning on a pre-trained ANN to discriminate PD from parkinsonism caused by other disorders. A support vector machine (SVM) trained using parameters of semi-quantitative measurements including specific binding ratio and asymmetry index was used for comparison. The predictive accuracy of the ANN classifier (86%) was higher than that of the support vector machine classifier (68%). The sensitivity and specificity of the ANN classifier in predicting PD were 81.8% and 88.6%, respectively.

Pahuja and Nagabhushan (2021) proposed techniques for the classification of PD over the PD voice dataset and reported an *Acc* of ⟨95.89, 72.31, 88.21⟩, *Sen* of ⟨93.75, 68.75, 91.67⟩ and *Spec* of ⟨96.59, 73.47, 77.55⟩, using ANN, KNN and SVM, respectively. Similarly, Varrecchia et al. (2021) proposed an automated diagnostic approach for the detection of the presence and severity of PD using ANNs and reported *Sen* of 72.37%, *Spec* of 73.13%, *AUC* of 77.0% over the Gait dataset. Chien, Hsu, Lee, Sung, and Lin (2021b) proposed an automated diagnostic approach for the classification of severity of PD. Firstly, striatal regions of the images were segmented using an active contour model and utilized as the data for transfer learning on a pre-trained ANN to distinguish

PD from other causes of parkinsonism, and reported *Acc* of 86%, *Sen* of 81.8%, *Spec* of 88.6%, *AUC* of 77.0% over the DAT-SPECT images. Shichkina, Irishina, Stanevich, and de Jesus Plasencia Salgueiro (2021) proposed an approach for the selection of the NN along with LSTM for the diagnosis of severity of PD patients and reported an *Acc* of 97.29% over the voice records. Yang et al. (2021) have introduced a classification technique for the diagnosis of PD by fusing neuroimaging and clinical information (CI) using novel stacking ensemble (SVM, RF, KNN, ANN, LR) model and reported an *Acc* of 96.88%, *Sens* of 95.0%, and *F1-score* of 97.44% using 10-fold CV. de Souza et al. (2021) used RBM and fuzzy Optimum path forest (OPF) on hand-drawn images to classify the PD vs CN and reported an *Acc* of 79.57% and *F1-score* of 70.33% using hold-out validation scheme.

3.2. CNN based architectures

Eskofier et al. (2016) suggested a deep learning-based approach to track motor symptoms' activity results within and outside the clinic over the dataset obtained from ten idiopathic patients with PD. Tagaris, Kollias, and Stafylopatis (2017) also proposed a PD detection framework based on deep neural networks. A cascaded CNN with a bidirectional long short-term memory (LSTM) network is proposed for learning intra-slice and inter-slice features from 2D DaT Scans. PD also affects gait of affected patients and can also act as a marker for the disease. To utilize gait-based features, Um et al. (2017) proposed a deep CNN for gait signals acquired from wearable sensors. A data augmentation scheme was also performed to increase the training sample size. 2D-CNN models do not take into account the 3D spatial nature of medical images. In order to learn more fine-grained features from 3D volumes, Martinez-Murcia et al. (2017) proposed a 3D-CNN for PD and SWEDD detection. Choi, Ha, Im, Paek, and Lee (2017) worked on the interpretation of the deep CNN model for PD diagnosis. The proposed seven-layer deep architecture produced a high accuracy for PD as well as SWEDD detection.

In order to identify PD, Zhang, Wang, Li, and Xu (2018) developed a deep CNN for spectrogram images. The spectrogram images were generated from voice signals for a mobile health framework. In order to utilize diverse features from both raw signals and their spectrograms, Zhao, Qi, Li, Dong, and Yu (2018) proposed a hybrid system consisting of parallel CNN and LSTM paths. Image spectrograms of gait signals were passed through the CNN path and sequential gait signals were passed through the LSTM path. The final classification is the result of an ensemble from the two parallel paths. Khojasteh et al. (2018) also proposed a CNN architecture for PD classification using voice signal spectrograms. Kim, Wit, and Thurston (2018) used a pretrained inception network to classify PD subjects using 2-D DaT Scan slices, thereby, utilizing transfer learning to boost the performance of the network. Adams, Yang, Rahmim, and Tang (2018) proposed a 3D CNN for motor function prediction using DaT SPECT images. Khatamino, Cantürk, and Özyılmaz (2018) proposed a deep network to detect PD from extracted signal images from handwriting signals (HWS). Wu et al. (2018) performed tensor decomposition to decompose 3D PET scans to 2D scans. The 2D scans were used to train and test a deep CNN for differential diagnosis of PD.

Chen, Shi, Peng, and Dai (2018) proposed an ensemble of stacked deep polynomial networks to detect PD from MRI scans. Esmaeilzadeh et al. (2018) designed an eight-layer deep CNN for differentiating PD subjects from HC subjects using 3D T1 weighted MR images. The developed model also leveraged the information conveyed from age and gender of the subjects. The addition of normalization procedures (batch normalization and group normalization) to the designed model increased the validation accuracy up to 100%. Pereira et al. (2018) proposed a novel methodology for detecting PD from the images of time series signals acquired during specific handwriting tasks by using a smartpen equipped with various sensors. The images were passed on to a deep CNN to extract complex features and classification. A distinct CNN is trained on each handwriting task, and the final ensemble results in a much better performance for identifying PD and HC subjects. Oh et al. (2018) proposed a thirteen-layer CNN for extraction of features from electroencephalography (EEG) signals using 1D convolutions. The proposed architecture uses stride pooling layers to decrease feature map size rapidly. The proposed CNN achieves better performance than the previous state-of-the-art techniques of PD prediction and diagnosis from EEG signals.

Vásquez-Correa et al. (2018) proposed a CNN architecture that leveraged information from multimodal data (speech, HWS and gait signals) for PD classification. A global average pooling is performed in the last layer of the CNN to generate multimodal features containing individual feature modality information, then passed on to an SVM classifier for final classification. Hwang et al. (2018) utilized the combination of convolutional autoencoder (CAE) and U-Net for reducing the noise in 3D PET images. A deconvolution layer is also used in this architecture for increasing feature map size. CVásquez-Correa, Arias-Vergara, Orozco-Arroyave, and Nöth (2018) proposed multitask learning (MTL) based deep learning technique for classification of PD and HC and also for gauging the Dysarthria (Roth, 2011) severity in PD subjects. The MTL provides better generalization results as compared to single-task CNN. Kim, Lee, et al. (2018) proposed a deep CNN for tremor severity classification from image spectrograms of gait signals acquired from wrist-wearable sensors.

Ortiz et al. (2019) used isosurfaces for feature extraction from high dimensional 3D SPECT images. Before using the isosurface techniques, spatial and intensity normalization are used to extract features and feed into two different 3D CNNs based on LeNet and AlexNet. A model interpretation technique was also employed through the use of saliency maps, to identify regions having more importance on the classification. Gil-Martín, Montero, and San-Segundo (2019) proposed a deep model utilizing the 1D-convolution operation to classify PD subjects using time-series handwriting signals (HWS). Kollia, Stafylopatis, and Kollias (2019) leveraged combined information from three consecutive MRI slices and the corresponding DaT Scan image. The slices were input to a pretrained CNN that acted as a feature extractor for a stacked gated recurrent unit (GRU) layer and used the CNN-RNN architecture to classify subjects in a hospital environment. Sivaranjini and Sujatha (2019) developed an AlexNet-like network to detect PD from 2D slices extracted from 3D MRI-T2 scans. Wodzinski, Skalski, Hemmerling, Orozco-Arroyave, and Nöth (2019) developed a CNN based

encoder for classification of PD data on image spectrograms generated from voice data. Xiao et al. (2019) proposed a DenseNet inspired architecture utilizing the dense connections for PD classification from Quantitative Susceptibility Mapping (QSM) images.

Sigcha et al. (2020) transformed gait signals acquired from an accelerometer into spectral representations to detect freezing of gait in PD subjects. Each subject's spectral representation was considered an input to a 1D-convolutional network stacked with a LSTM layer for modelling the temporal sequence of signals. The combination of CNN with LSTM outperformed the other models due to its rich feature extraction and temporal sequence modelling capabilities. Bouza et al. (2020) proposed theory and application for a novel model named manifold-valued convolution network (MVC-Net), which is a generalized model for dealing with manifold-valued images that are usually found in the biomedical imaging domain. MVC-Net was applied to the classification of PD and HC subjects using diffusion MRI. Oktay and Kocer (2020) used a combination of 1D-convolutional neural network and LSTM on the hand tremor and postural tremor signals acquired from a leap motion controller (LMC). The CNN extracts rich feature representations from the tremor signals, whereas the LSTM layer accumulates the time series sequential signals. This combination provides better performance for classification of essential tremor (ET) and parkinsonian tremor (PT).

Naseer et al. (2020) studied different deep transfer learning frameworks of AlexNet for classification of small PD dataset. AlexNet weights were frozen in one setting, whereas, on the other, the weights were allowed to adapt to the new dataset. A linear SVM classifier was used for classification. Pianpanit et al. (2020) tested four different 3D CNN architectures on SPECT imaging dataset for PD diagnosis. The different models are also tested for interpreting the predictions by CNN. Guided backpropagation outperformed other models with the best accuracy as well as interpretation results. Zanca, Rufa, Canessa, and Sabatini (2020) utilize raw pupil size registration time-series signals and 1D-CNNs to diagnose PD and a data-driven batch normalization is also used for normalizing the inputs before convolution. Golshan, Hebb, and Mahoor (2020) introduced the idea of utilizing subthalamic nuclei local field potential (STN-LFP) signals obtained through deep brain stimulation and CNNs for recognizing human behavioural activities. Pang et al. (2020) utilized the pre-trained old pose model to identify Bradykinesia by detecting tremor in hand movements. El Maachi, Bilodeau, and Bouachir (2020) proposed a deep model leveraging an ensemble of 1D-CNN on the vertical ground force (VGRF) time-series signals for PD detection and severity prediction.

Adams, Rahmim, and Tang (2021) proposed a deep learning-based technique for predicting motor function using longitudinal clinical information (CI) with DaT SPECT images. The motor function prediction problem is first transformed into a category prediction task. Then, a custom 3D CNN is used to learn and predict the new task. Bikias, Iakovakis, Hadjidimitriou, Charisis, and Hadjileontiadis (2021) proposed a light CNN to learn FoG in PD patients using movement data acquired from accelerometer and gyroscope. Vyas et al. (2021) compared the performance of 2D and 3D CNNs for PD classification from MRI data. Ismail Canturk et al. analysed dynamic spiral test (DST) and static spiral test (SST) signals in a two stage framework using pretrained deep CNNs. The pretrained deep CNNs performed feature extraction. The extracted features were evaluated in the first stage. In the next stage, the best features were then passed onto another CNN for feature extraction. The extracted features were classified using SVM and k-nearest neighbour (KNN) algorithms. Bhan et al. performed PD classification using a light CNN and MRI data. In order to learn novel patterns from EEG data for PD diagnosis, Loh, Ooi, et al. (2021) converted EEG data acquired from PD patients to image spectrograms. A 2D CNN was then trained on the spectrograms for learning to perform PD classification. Conventional CNNs can be very computationally intensive due to the presence of millions of parameters, and can also take a large number of iterations to learn properly. Majda et al. also used image spectrograms to compare between the PD prediction capabilities of SVM and CNN. In order to make a more efficient CNN for PD diagnosis, Yadav (2021) proposed a Bayesian deep CNN to learn the task of PD classification from functional magnetic resonance imaging (fMRI) data.

Heidarivincheh et al. (2021) addressed the problem of missing data modalities for PD diagnosis by utilizing the generation capabilities of variational autoencoders (VAEs). The proposed method can utilize the diversity present in multiple modalities as well as deal with the scenario when one of the modality is not present. Ashfaque Mostafa, Soltaninejad, McIsaac, and Cheng (2021) proposed an ensemble of multiple deep neural networks trained on different feature views for FoG prediction. The results were based on stacked ensemble, majority voting and model averaging. Shin et al. used the YoloV3 (Redmon & Farhadi, 2018) network for assessing nigrostriatal degeneration.

3.3. RNN based architectures

Lacy, Smith, and Lones (2018) studied the efficacy of echo state networks (ESN) in PD diagnosis. The ESNs work on time-series movement signals. The study suggests the use of ESN for high dimensional time-series signals may prove beneficial.Gallicchio, Micheli, and Pedrelli (2018) proposed a deep ESN for PD detection using time-series handwriting signals. Pereira et al. (2017) used features generated by a restricted Boltzmann machine (RBM) to feed into three different classifiers. The SVM with the RBF kernel combined with RBM generated features were most useful for PD diagnosis using signals from handwriting examinations.

Masiala, Huijbers, and Atzmueller (2019) proposed a deep LSTM network for detection of freezing of gait (FoG) in PD subjects. Tang, Hoang, Chui, Lim, and CH (2019) used an LSTM to classify gait signals from a wearable device. Ribeiro et al. (2019) utilized the gated recurrent unit (GRU) network to model the temporal information conveyed by tremor signals from handwriting exams. A novel bag of signals (BoS) method is used for compact computing representations of signals, which are then used by the RNN for classification. Afonso et al. (2019) proposed a recurrence plot-based technique to convert the time series signals to image domain and then pass the images to a CNN for final classification PD vs HC. The method found that low-resolution images do not give better results as compared to the high-resolution images.

Ashour et al. (2020) designed a patient-dependent two-layer LSTM network to detect freezing of gait sequences from gait signals acquired through an accelerometer tied to the subjects' hip knee and ankle and passed the gait signals through a combination

of discrete wavelet transform (DWT) and fast Fourier transform (FFT) to extract features for the network. Alfaro-Ponce and Chairez (2020) experimented with gait signals and different types of continuous-valued dynamic recurrent neural networks. An implementation on 32-bit microcontroller was also performed. The RNN outperformed conventional ANN with a final classification accuracy of 95%. Lin et al. (2020) proposed periodic motion network (PMNet), a deep encoder–decoder LSTM architecture for bradykinesia recognition from a single video of each subject. Szumilas, Lewenstein, Ślubowska, Szlufik, and Koziorowski (2020) presented an ensemble of randomly initialized echo state networks (ESN) for the quantification of tremor signals of PD patients. Kamran, Naz, Razzak, and Imran (2021) proposed a model to detect the PD from handwriting images using applied deep transfer learning. The usage of transfer learning helped the model to the patterns in the PD dataset more quickly and accurately. Instead of performing manual feature selection of PD diagnosis from speech data, Demir, Sengur, Ari, Siddique, and Alswaitti (2021) converted input features to images and utilized convolutional layers to learn the appropriate features. The learnt features were formed into a sequence which was learnt by an LSTM layer. Sivakumar, Christinal, and Jebasingh (2021) used a combination of a pretrained CNN with an LSTM layer to perform PD diagnosis using hand drawn images. Thummikarat and Chongstitvatana (Thummikarat & Chongstitvatana, 2021) utilized features extracted data of PD subjects from keyboard, gyroscope and an accelerometer for PD classification. The extracted sequential features were passed through a deep LSTM network for final PD classification.

3.4. Miscellaneous deep architectures

Zhang (2017) worked on verifying smartphone-based PD diagnostic systems' effectiveness and used a combination of SAE and K-nearest neighbors (KNN) on speech data acquired from a Smartphone to detect voice impairment patterns that are a common symptom of PD patients. The proposed approach performed better classification of the PD subjects while being independent of the number of hidden neurons and the layer depth.

Nilashi, Ibrahim, Ahmadi, Shahmoradi, and Farahmand (2018) combined self-organizing maps (SOM), nonlinear iterative partial least squares (NIPALS) and immune support vector regression (ISVR) for predicting total unified PD rating scale (UPDRS) and motor UPDRS scores. The SOM was used for clustering the data points, whereas the NIPALS was used for dimensionality reduction. Then the ISVR was used for regression and prediction of UPDRS scores.Gong et al. (2018) proposed a novel deep large margin distribution machine (DLLMD) for overcoming the problem of kernel selection in LMD. The DLLMD was used for PD diagnosis. Passos et al. (2018) used pre-trained Residual networks (He, Zhang, Ren, & Sun, 2016) as feature extractors for handwriting images and evaluated extracted features on different classifiers like OPF, SVM and NB classifier.

Shi et al. (2018) proposed a cascaded multi-column random vector functional link (RVFL) classifier to detect PD from MRI scans and leverage some privileged information during training but not during testing. Kadam and Jadhav (2019) proposed a sparse autoencoder (SAE) based multi-path deep network ensemble for classification of PD subjects using features extracted from speech data. Peng, Guan, and Shang (2019) used node2vec (Grover & Leskovec, 2016) model to create network-based embedding of genetic data. The embedding dimension was then reduced using an autoencoder, and the final classification was done using the SVM model. Qin, Jiang, Chen, Hu, and Ma (2019) proposed a lightweight deep 1D-CNN for detection of tremor severity from surface electromyography (sEMG) signals.

Akyol (2020) compared the performances of ELM with DNN for classification of PD and HC subjects using speech data. The classification was based on the performance evaluation from neuron growing and pruning approaches. DNN with stochastic gradient descent optimizer (SGD) and rectified linear unit (ReLU) activation outperformed the ELM for this task. OOlivares et al. (2020b) optimized the architecture of a one-layer ELM using the nature-inspired BAT algorithm on PD and HC subjects' speech features acquired from the UCI-ML repository. In order to utilize diverse predictions from various classifiers, Kakulapati and Santhosh (2020) introduced an ensemble learning based Random Multimodal Deep Learning (RMDL) model. RMDL provided more robust predictions for PD diagnosis. Salmanpour et al. (2021) have proposed PD Classification using hybrid ML systems (HMLSs) such as Local Linear Model Trees Algorithm (LOLIMOT), RBF, RNN, MLP-B, and RFA, by making significant usage of information retrieval over SPECT images with radiomics features and reported UPDRS III: (22.82, 24.22, 25.41); MMSE: (31.75, 32.6, 33.72) for PD subtypes — Mild, Moderate, and Severe, respectively. Masud et al. proposed a combination of an adaptive crow-search algorithm (ACSA) along with a deep fully connected stacked autoencoder (DFSAE) for PD diagnosis using voice data. Shaban et al. also features extracted from EGG data to train a fully connected feed forward deep network for classification of PD.

4. Discussion

Table 2 provides detailed information about the model architecture used in the research papers reported in Table 1. The 'problem type' column refers to the type of problem that the corresponding research dealt with. It also gives an idea about the type of data (time-series, feature-based or image-based) used for training the model. To get further insights into the model architecture, we divided the number of hidden layers to three groups — 'Initial', 'Middle' and 'Last'. The 'Initial' column represents the layer immediately next to the input. The 'Last' layer represents the layer immediately before the output for ANNs. The 'Last' layer for CNN based and other deep architecture represents the layer before the dense layers. The Middle column represents the layers in between the 'Initial' and 'Last' layers. The 'regularization' column gives details about various regularization techniques applied to convolution and dense layers. The 'learning rate' column represents the initial learning rate used in the model and the learning rate annealing techniques used. The 'hyperparameter tuning' column gives information about the set of optimal hyperparameters for the said learning algorithm used for training the model. The 'normalization' column depicts the normalization techniques used in the model for the convolution, input and dense layers. The 'activation' column represents the activation function used in the intermediate and the last layers.

4.1. Discussion about models covered in this study

We discuss the various types of deep and shallow neural networks covered in this work:

2D-CNN models: Naseer et al. (2020) and Xue & Xin Xue and Xin (2019) use a very large initial kernel. The former uses the transfer learning approach from the AlexNet model (Krizhevsky, Sutskever, & Hinton, 2012) pretrained on the ImageNet dataset (Deng et al., 2009), whereas, the latter trains the model from scratch. The large kernel size in these models helps in reducing the input dimension. However, a downsized of using large-sized kernels is the increase in the number of trainable parameters of the model, which further limits the model's depth. Naseer et al. (2020) also utilized momentum and a learning rate drop factor, making the model reach relatively better local optima. To further improve the model generalization, rotation and flipping data, augmentation routines were also performed. Golshan et al. (2020) also used 2D convolution layers. The convolution layers were also followed by a batch normalization (BN) layer. The BN layer is used after the nonlinearity. However, the BN layer's intended use is before the application of activation function (Ioffe & Szegedy, 2015).

Xue and Xin (2019) on the other hand, the leveraged sparse structure within the CNN for boosting the model performance. However, the effect of momentum and learning rate annealing on the proposed model was not studied. In Naseer et al. (2020), there were multiple handwriting images extracted from a single subject. The whole set of images were divided into training and testing sets. This may be a possible cause of data leakage as there is no subject-wise separation of data. Hwang et al. (2018) also used smaller convolution kernels and deconvolution operations to reconstruct the input.

Pereira et al. (2018) and Afonso et al. (2019) also used the AlexNet model. Pereira et al. (2018) use the ensemble of models trained on images from handwritten examinations. The usage of ensemble strategy results in a more generalized model (Ditterrich, 1997). Whereas, Afonso et al. (2019) used the recurrence plots for training the deep network. Sivaranjini and Sujatha (2019) used the transfer learning by fine-tuning the pretrained AlexNet. This method also is prone to data leakage, as no information on subject-wise training and testing data is given. Zhang et al. (2018) also used a similar-sized kernel for training with spectrograms generated from speech data. Ismail Canturk performed feature extraction using pretrained AlexNet and GoogleNet architectures. The extracted features were passed on to SVM and k-NN algorithms for final classification. Loh et al. performed convolution with a larger sized initial kernel (5×5) and a smaller sized convolution kernel of size 3×3. A constraint in weights of the kernels was also fixed in order to keep the weights within a specific range. Dropout is used in the dense as well as convolutional layers.

Notations for Table 2: We follow different notations for describing CNN, ANN and RNN based architectures:

CNN based architectures. For CNN based architectures, each of the initial, middle and last columns have to be read from the left to right. There can be multiple layers in any of the initial, middle of last categories. Hence, the configuration of each layer is separated by a comma ','. We consider the combination of the convolution and pooling layer as one layer. The convolution and pooling operations of a layer are separated by the plus sign '+'. For the 1D convolution operation, the acronym *1D conv* is used followed by (M × N), wherein, 'M' is the size of the kernel and 'N' is the number of channels. For the 2D convolution operation, the size of each kernel tensor is described by the format 'H × W × C', wherein, 'H' represents the height of the kernel, 'W' represents the width of the kernel and 'C' represents the number of channels. For the 3D convolution operation, the format of 'A^3×B' is followed, wherein, 'A' is the dimension of the cube size kernel and 'B' is the number of channels.

RNN based architectures. For the RNN layers, we use the name of the recurrent layer followed by '(Y)', wherein, (Y) denotes the number of hidden states in the network.

ANN based architectures. For the ANN based architectures, we use the format of I(J), wherein, 'I' represents the number of layers and 'J' represents the number of nodes in each layer. Each different layer is separated by a comma.

3D-CNN models: Choi et al. (2017) and Pianpanit et al. (2020) use strided 3D convolutions and strided max-pooling layers. The combination of strided convolutions and pooling layers result in a greater extent of input dimensionality reduction. However, the latter focused on model interpretability. Ortiz et al. (2019) also used 3D convolutional layers. However, the initial convolution was considered to be a larger size as compared to Choi et al. (2017). The larger size kernel can help in reducing the input dimension to a larger extent. However, it may cause the network to overlook minute details in the input. vadams2018prediction also used a kernel size configuration similar to Ortiz et al. (2019), the difference being that the former used a combination of strided convolution and pooling operations. Esmaeilzadeh et al. (2018) used smaller kernels for the convolution layer with strided max-pooling operation. However, the model also focuses heavily on reducing overfitting with bias, kernel regularization and normalization techniques. Data augmentation is also used, which further enhances the generalization capability of the model. The exponential LR decay is also used, which helps in the better tuning of weights. Adams et al. (2021) used a larger size initial convolution kernel ($7 \times 7 \times 7$) to extract features from a larger region. A larger stride value of 4 is also used for reducing the feature dimension for the subsequent layer. A strided max-pooling layer is further used to drastically reduce the size of the feature map. A gradually reducing kernel dimension of $5 \times 5 \times 5$ and $3 \times 3 \times 3$ are used for the subsequent convolution kernels. The network contains two similar parallel branches. Vyas et al. also used a convolution kernel of size $3 \times 3 \times 3$ along with strided max-pooling layers. Dropout layers are also used in dense layers for further regularizing the network. Samir Yadav performed dropout before passing the feature map to the convolutional layers. Dropout is also used after convolution operation. The network is a lightweight network guided by Bayesian statistical principles. Heidarivincheh et al. (2021) utilized 3D convolution on videos of PD subjects performing tasks. A VAE was proposed that minimized the differences in the latent space between videos and accelerometer signals. This enabled the model to be agnostic to a missing modality. The proposed encoder and decoder followed a symmetric structure.

Table 1
Comparison of research on classification of PD based on Architecture of the models (GenInfo-1)

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
1	Gharehcho-pogh and Mohammadi (2013)	2013	24 PD vs 8 CN	Speech	–	ANN	Hold-out	93.22	–	–	–	–
2	Hossen (2013)	2013	39 PD vs 41 ET	Acclrm and EMG signals	wavelet	ANN	Hold-out	87.5	–	–	–	–
3	Babu and Suresh (2013)	2013	50 PD vs 22 CN	Gene expression	ICA	PBL-McRBFN (ANN)	Hold-out	97.17	–	–	–	–
4	Mandal and Sairam (2013)	2013	23 PD vs 8 CN	CI and Speech	–	LR, BayNet, SVM, ANN, Boosting, Ensembles	LOOCV	97.74	–	–	97.8	–
5	Hosseini and Makki (2013)	2013	16 PD	Tremor	PCA	Autoassoc-iative NN	–	97.3	97.5	97.1	97.3	–
6	Hariharan et al. (2014)	2014	24 PD vs 8 CN	Speech	–	PNN, GRNN	10-fold	100, 100	–	–	–	–
7	Khorasani and Daliri (2014)	2014	16 PD vs 15 CN	Gait	–	HMM Net	LOOCV	90.32	93.33	87.5	–	–
8	Sharma and Giri (2014)	2014	23 PD vs 8 CN	Speech	–	MLP (NN), KNN, SVM	–	82.35 (MLP)	84.61	75	–	–
9	Beheshti, Shamsuddin, Beheshti, and Yuhaniz (2014)	2014	48 PD vs 149 CN	Speech	–	CAPSO-MLP, PSO-MLP, GSA-MLP, ICA-MLP	–	Mean –92.56, 92.56, 82.05, 92.31	Mean: 93.96, 93.96, 82.05, 93.94	Mean-85.48, 85.00, 0.00, 83.33	0.912, 0.898, 0.596, 0.865	–
10	Islam et al. (2014)	2014	23 PD vs 8 CN	Speech	–	RT, SVM, FB-ANN	10-fold	97.73	98.6	93.62	RoC>0.9	–
11	Olanrewaju et al. (2014)	2014	23 PD vs 8 CN	Speech	–	MLPFNN	Hold-out	80	83	63	–	–
12	Joo et al. (2014)	2014	20 PD	Gait	–	ANN	5-fold	–	–	–	–	CC= 97.8
13	Shaikh (2014)	2014	23 PD vs 8 CN	Speech	–	ANN, DT, NB	10-fold	ANN- 90.76	85.41	92.51	–	–
14	Mandal and Sairam (2014)	2014	–	CI	–	Various ML algorithms including ANN	–	ANN- 98.2759	–	–	–	
15	Froelich et al. (2014)	2014	100 Sample	–	–	kNN, FCM, NB , ANN, FRBS, SVM	10-fold	ANN-0.75	–	–	–	–
16	Davidson et al. (2015)	2015	–	–	–	–	–	–	–	–	–	–

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
17	Peker et al. (2015)	2015	23 PD vs 8 CN	Speech	mRMR, Fisher score, Chi-square, SFS, SFFS, and ReliefF	CVANN	10-fold	98.12	–	–	–	–
18	Hirschauer et al. (2015)	2015	189 PD vs 62 SWEDD vs 415 CN	CI	–	EPNN	Hold-out	98.6 (PD vs CN)	–	–	–	–
19	Saloni et al. (2015)	2015	23 PD	CI	–	SVM, ANN	Hold-out	87.70 (ANN)	94.5	66.6	–	–
20	Er et al. (2016)	2016	197 PD	CI	–	PNN	10-fold	96.5	–	–	–	–
21	Hlavica et al. (2016)	2016	–	Speech	–	ANN	Hold-out	–	–	–	–	–
22	Eskofier et al. (2016)	2016	–	–	–	CNN	Hold-out	90.9	–	–	–	–
23	Jane et al. (2016)	2016	93 PD vs 73 CN	Gait	–	Q-BTDNN	10-fold	91.49, 92.19 and 90.91	–	–	–	–
24	Khan et al. (2016)	2016	23 PD vs 8 CN	Speech	WT	WNN	10-fold	92.93	–	–	–	–
25	Al-Fatlawi et al. (2016)	2016	24 PD vs 8 CN	Speech	–	DBN	Hold-out	94	–	–	–	–
26	Pereira et al. (2016)	2016	14 PD vs 21 CN	HWI	–	CNN	Hold-out	97.42	–	–	–	–
27	Turner et al. (2016)	2016	25 PD	Gait	–	epiNet	Hold-out	70	–	–	–	==
28	Del Din et al. (2015)	2016	30 PD vs 30 CN	Gait	–	–	–	–	–	–	–	–
29	Abiyev and Abizade (2016)	2016	23 PD vs 8 CN	Speech	–	FCN	10-fold	100	–	–	–	–
30	Benba et al. (2016)	2016	20 PD vs 20 CN	Speech	–	MLP	10-fold	100	–	–	–	–
31	Sivanesan et al. (2016)	2016	–	Visual stimulus, and DaT Scan	–	MP-NN	Hold-out	93	–	–	–	–
32	Gürüler (2017)	2016	23 PD vs 8 CN	Speech	–	KCFW + CVANN	10-fold	99.52	–	–	–	–
33	Zia et al. (2016)	2016	3 PD	Gait	–	NN	Hold-out	89	–	–	–	–
34	Challa et al. (2016)	2016	402 PD vs 301 CN	DaT Scan	–	MLP-ANN	Hold-out	95.45	–	–	–	–
35	Tagaris et al. (2017)	2017	100 PD vs 40 CN	–	–	CNN + RNN	Hold-out	74	–	–	–	–

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
36	Choi et al. (2017)	2017	431 PD vs 193 CN vs 77 SWEDD	SPECT	–	CNN	Hold-out	98.8	98.6	100	–	–
37	Zhang (2017)	2017	23 PD vs 8 CN; 20 PD vs 20 CN	Speech	TFF	SAE + KNN	10-fold	98.01	93.06	95.65	–	–
38	Pereira et al. (2017)	2017	74 PD vs 18 CN	HWS	–	RBM	–	83.12 ± 2.55	–	–	–	–
39	Nilashi et al. (2018)	2017	42 PD	Speech	–	SOM + NIPALS + ISVR	–		–	–	–	Total UPDRS: MAE=0.4656, R^2=0.885; Motor-UPDRS: MAE =0.4967, R^2=0.868
40	Gómez-Vilda et al. (2017)	2017	M: 53 PD vs 26 CN F: 38 PD vs 25 CN	Speech	–	RLSFN	LOOCV	99.45 (M) 99.42(F)	99.46(M) 99.42 (F)	99.44 (M) 99.41 (F)	99.47 (M) 99.45 (F)	–
41	Nikandish and Kheirkhah (2017)	2017	50 PD vs 41 CN	Gait	MLP	RF; SVM; KNN	–	78	–	–	–	–
42	Um et al. (2017)	2017	30 PD	WSS	–	CNN	5-fold	86.88	–	–	–	–
43	Zhang et al. (2018)	2018	500 PD vs 500 CN	Speech	STFT	CNN	Hold-out	90.45 ± 1.71	–	–	–	–
44	Passos et al. (2018)	2018	296 PD vs 72 CN	HWS	Resnet50	PCA + SVM/OPF/NB	5-fold	96.31	–	–	–	–
45	Shi et al. (2018)	2018	TCS: (76 PD vs 77 CN); PPMI: (68 PD vs 103 CN)	–	–	CMCRVFL	–	81.93 81.87	80.25 82.68	83.50 79.83	–	–
46	Zhao et al. (2018)	2018	93 PD vs 73 CN	Gait	–	CNN + LSTM	–	98.6	–	–	–	
47	Khojasteh et al. (2018)	2018	41 PD vs 40 CN	Speech	–	CNN	Hold-out	75.7	–	–	–	–
48	Kim, Wit, and Thurston (2018)	2018	54 PD vs 54 CN	SPECT	–	Inception-V3 (Pre-trained)	Holdout	–	96.3	66.7	87	81.3 (PPV)
49	Rumman et al. (2018)	2018	100 PD vs 100 CN	SPECT	–	ANN	–	94	100	88	–	–

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
50	Alaskar and Hussain (2018)	2018	14 PD vs 15 CN	Gait	–	MLP	–	91.18	–		–	–
51	Adams et al. (2018)	2018	293 PD	SPECT	–	CNN	10-fold	83.3	–	–	–	–
52	Khatamino et al. (2018)	2018	57 PD vs 15 CN	HWS	–	CNN	LOOCV	88	–	–	–	–
53	Wu et al. (2018)	2018	136 IPD vs 91 MSA vs 30 PSP	FDG-PET	TD	CNN	5-fold		98.9	82.5		96.1 (PPV)
54	Wroge et al. (2018)	2018	–	Speech	–	ANN	Stratified 10-fold	86 ± 0.01	(82 ± 0.02)			75 ± 0.03 (Precision), 78 ± 0.03 (F1-score)
55	Gallicchio et al. (2018)	2018	–	HWS	–	Ensemble Deep-ESN	3-fold	90	–	–	–	–
56	Chen et al. (2018)	2018	–	–	–	Stacked DPN	–	87.18	86.9	87.27	–	–
57	Esmaeilzadeh et al. (2018)	2018	452 PD vs 204 CN	MRI-T	–	CNN	Hold-out	100	–	–	–	–
58	Pereira et al. (2018)	2018	74 PD vs 18 CN	HWS	–	CNN	10 fold	–	99.3	70.51	–	–
59	Oh et al. (2018)	2018	20 PD vs 20 CN	EEG	–	CNN	10-fold	88.25	84.71	91.77	–	–
60	Vásquez-Correa et al. (2018)	2018	44 PD vs 40 CN	Speech, HWS, Gait	STFT	CNN	10-fold	97.26	–	–	–	–
61	Hwang et al. (2018)	2018	40 PD	PET	–	CAE + U-Net	5-fold	–	–	–	–	–
62	Grover et al. (2018)	2018	5875 PD	Speech	–	DNN (MLP)	Hold-out	81.667	–	–	–	–
63	Vásquez-Correa et al. (2018)	2018	50 PD vs 50 CN	Speech	–	CNN + MTL	Hold-out	80	–	–	–	–
64	Kim, Lee, et al. (2018)	2018	92 PD vs 95 CN	Tremor	FFT	CNN	Hold-out	85	–	–	–	kappa=85, CC=93, rmse=35
65	Gong et al. (2018)	2018	68 PD vs 103 CN	MRI	–	LMD	5-fold	81.70 ± 1.77	82.92 ± 3.29	80.02 ± 5.28	–	–
66	Lacy et al. (2018)	2018	107 PD vs 70 CN	MS	–	ESN	10-fold	–	–	–	80.2	–
67	Parisi et al. (2018)	2018	20 PD vs 20 CN	Speech	–	ANN + SVM	–	100	100	100	–	–
68	Khan et al. (2018)	2018	43 PD vs 28 CN	Speech	–	ENN	10-fold	100	–	–	–	–

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Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
69	Martinez-Murcia et al. (2017)	2018	158 PD vs 32 SWEDD vs 111 CN	SPECT	–	3D-CNN	Hold-out	PD vs CN: 95.5 ± 0.44	PD vs CN: 96.2 ± 0.051	–	–	–
70	Wodzinski et al. (2019)	2019	50 PD vs 50 CN	Speech	–	CNN	10-fold	91.7	–	–	93	–
71	Xiao et al. (2019)	2019	87 PD vs 53 CN	QSMI	–	CNN	Hold-out	–	–	–	96	–
72	Peng et al. (2019)	2019	178 PD	Genetic dataset	–	Node2vec + AE + SVM	–	70.56(ROC)	–	–	–	–
73	Qin et al. (2019)	2019	157 PD	Tremor	–	CNN	–	90.55	–	–	–	–
74	Masiala et al. (2019)	2019	10 PD	Gait	–	LSTM	–	–	81	90	93	–
75	Tang et al. (2019)	2019	–	Gait	–	RNN	–	85.11	85 (Recall)	–	–	87 (Precision)
76	Kadam and Jadhav (2019)	2019	23 PD vs 8 CN	Speech	–	Sp-AE	–	90	–	–	–	–
77	Mikos et al. (2019)	2019	63 PD	Gait	–	ANN	–	–	95.6	90.2	–	
78	Afonso et al. (2019)	2019	224 PD vs 84 CN	HWS	RP	CNN	Hold-out	88.04	97.86	95.54	–	–
79	Ribeiro et al. (2019)	2019	54 PD vs 20 CN	HWS	BoS	GRU	5-fold	95.24	–	–	–	
80	Johri and Tripathi (2019)	2019	(138 PD vs 57 CN), (93 PD vs 73CN)	Speech, Gait	ImgSpect	CNN	–	89.15, 88.17	–	–	–	–
81	Naseer et al. (2020)	2019	37 PD vs 38 CN	HWI	–	AlexNet + SVM	4-fold	98	–	–	–	–
82	Xue and Xin (2019)	2019	6000 (All)	HWI		CNN + RVSM	–	99	–	–	–	–
83	Ortiz et al. (2019)	2019	158 PD vs 111 CN	SPECT	Isosurface	CNN	10-fold	95	95.5	94.8	97	–
84	Vital et al. (2020)	2019	46 PD vs 36 Non-PD	Speech	–	ANN	–	100	–	–	–	–
85	Gil-Martín et al. (2019)	2019	62 PD vs 15CN	HWS	FFT	CNN	5-fold	96.5	–	99.7	–	–
86	Kollia et al. (2019)	2019	55PD vs 23 non PD	MRI and DaT Scan	–	CNN-RNN	–	98	–	–	–	–
87	Ali et al. (2019)	2019	20 PD vs 20 CN	Speech	–	GA + ANN	LOOCV	82.14	–	–	–	–
88	Mostafa et al. (2019)	2019	23 PD vs 8 CN	Speech	–	ANN, DT, SVM, NB, RF	10-fold	87.75 (ANN)	87.8	–	–	88.2 (Precision)

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
89	Sivaranjini and Sujatha (2019)	2019	2820 CN vs 3296 PD	MRI-T2	–	CNN	Hold-out	–	89.3	88.4	91.8	–
90	Golshan et al. (2020)	2020	10 PD	LFP signals and Speech	ImgSpect	–	10-fold	88	–	–	–	–
91	Pang et al. (2020)	2020	5 PD vs 22 CN	HFM Signal	DWT	CNN	–	–	–	–	–	0.768 (DBI-cluster separability score)
92	Alfaro-Ponce and Chairez (2020)	2020	93 PD vs 73 CN	Gait	–	RNN	5-fold	95	–	–	–	–
93	Lin et al. (2020)	2020	94 PD vs 83 CN	HMV	–	CNN	10-fold	77.78	82.35 (precision)	–	–	77.78
94	El Maachi et al. (2020)	2020	93 PD vs 73 CN	VGRF	–	1D-CNN	10-fold	98.7	–	–	–	–
95	Szumilas et al. (2020)	2020	64 PD	Tremor	STFT	RNN	5-fold	–	–	–	–	0.629 (CORR with UPDRS)
96	Kaur, Aggarwal, and Rani (2020a)	2020	10 PD	Gait	–	CNN-RNN	Hold-out	82.3	–	–	–	–
97	Li et al. (2020)	2020	10 PD	Gait	–	CNN-RNN	10-fold	99	99	99	–	–
98	Wingate, Kollia, Bidaut, and Kollias (2020)	2020	423 PD vs 196 CN	MRI + DaT Scan	–	CNN + LSTM	–	98	–	–	–	
99	Wang, Lee, Harrou, and Sun (2020)	2020	401 PD vs 183 CN	Clinical Info.	–	FFNN	5-fold	96.45	–	–	–	
100	Kaur, Aggarwal, and Rani (2020b)	2020	188 PD	Speech	–	FFNN	10-fold	91.69	–	–	–	
101	Choi, Kim, Yoon, Lee, and Lee (2020)	2020	49 PD vs 13 PDD	FDG-PET	–	3D CNN	Hold-out	–	–	–	81	–
102	Zhang, Deng, Li, Albin, and Guan (2020)	2020	2148 CN vs 656 PD	Gait	–	1D CNN	5-fold	–	–	–	–	87 (ROC)
103	Ashour et al. (2020)	2020	10 PD	Gait	DWT + FFT	LSTM	–	83.83	–	–	–	

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
104	Akyol (2020)	2020	564 PD vs 192 CN	Speech	–	DNN		95.15	–	–	–	
105	Olivares et al. (2020b)	2020	188 PD vs 64 CN	Speech	–	ELM	–	96.74	–	–	–	
106	Sigcha et al. (2020)	2020	21 PD	Gait	FFT	CNN + LSTM	LOOCV	93.9	–	–	–	–
107	Bouza et al. (2020)	2020	355 PD vs 356 CN	MRI		MVC-Net	10-fold	94.9	–	–	–	–
108	Oktay and Kocer (2020)	2020	23 PT vs 17 ET	Tremor	–	CNN + LSTM	LOOCV	90	–	–	–	–
109	Shahtalebi et al. (2020)	2020	81 PD vs 81 ET	Tremor	–	BGRU	–		–	–	–	0.00104 (MSE)
110	Pianpanit et al. (2020)	2020	48 PD vs 159 CN	SPECT	–	3D CNN	10-fold	96.87 ± 2.13	97.10 ± 2.35	97.89 ± 3.6	–	–
111	Zanca et al. (2020)	2020	15 PD vs 21 CN	PSR	–	1D-CNN	5-fold	83.67 ± 0.29	–	–	–	–
112	Rasheed et al. (2020)	2020	8 CN vs 23 PD	Speech	PCA	ANN	500 iteration	97.50	–	–	–	–
113	Wanjale et al. (2020)	2020	42 PD	Speech	–	ANN	Hold-out	83.56	–	–	–	–
114	Chakraborty et al. (2020)	2020	203 CN vs 66 PS vs 637 PD	MRI	–	ANN	5-fold	95.3	95.41	–	–	F1S-94
115	Veeraragavan et al. (2020)	2020	93 PD	MRI	–	ANN	k-fold	97.41	97.70	97.05		
116	Olivares et al. (2020a)	2020	64 CN vs 124 PD	Speech	Spectrograms	Bat algorithm + ELM	Hold-out	96.34	–	–		
117	Huang et al. (2020)	2020	196 PD	SPECT	– PCA, MPCA, Isat	MLP RF Deep CNNs	Hold-out	49.5 55.5 64.9	– –	– –	–	–
118	Kamran et al. (2021)	2020	193 CN vs 244 PD	Handwriting	–	CNN	Hold-out	99.22	–	–		
119	Chien et al. (2021a)	2020	193 CN vs 244 PD	DAT-SPECT	–	ANN	Hold-out	99.22	81.8	88.6	–	–
120	Mohammed, He, and Lin (2020)	2021	1359 CN vs 1364 PD	SPECT	–	2D CNN	10-fold	99.34	99.04	99.63	–	–
121	Yasaka et al. (2021)	2021	115 CN vs 115 PD	MRI	–	2D CNN	5-fold	–	–	–	89.5	–
122	Quan, Ren, and Luo (2021)	2021	15 CN vs 30 PD	Speech	–	RNN	10-fold	–	–	–	–	–

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
123	Leung, Rowe, Pomper, and Du (2021)	2021	198 PD	DaT Scan	LSTM	CNN, Ensemble	Hold-out	–	–	–	–	0.84 (PCC)
124	Adams et al. (2021)	2021	252 PD	DaT SPECT	–	CNN	10-fold	–	–	–	–	6.0 ± 4.8 (UPDRS_III)
125	Demir et al. (2021)	2021	188 PD vs 64 CN	Voice records	–	Deep LSTM	Hold-out	94.27	96.0	96.0	–	93.0 (F-score)
126	Pahuja and Nagabhushan (2021)	2021	23 PD vs 8 CN	Voice records	–	ANN	–	95.89	93.75	96.59	–	–
127	Varrecchia et al. (2021)	2021	76 PD vs 67 CN	Gait	PCA	ANN	Hold-out	–	72.37	73.13	77.0	–
128	Chien et al. (2021b)	2021	379 PD	DaT SPECT	–	ANN	10-fold	86.6	81.8	88.6	–	–
129	Shichkina et al. (2021)	2021	28 PD	Voice records	GA	GA + NN + LSTM	–	97.29	–	–	–	–
130	Yang et al. (2021)	2021	65 PD vs 36 CN	MRI + CI	–	Ensemble (SVM, RF, KNN, ANN, LR)	10-fold	96.88	95.0	–	–	97.44 (F1- score)
131	de Souza et al. (2021)	2021	420 PD vs 212 CN (Sample/ Instances)	Hand-drawn Images	RBM	Fuzzy OPF	Hold-out	79.57	–	–	–	70.33 F1-score
132	Salmanpour et al. (2021)	2021	885 PD vs 163 CN	SPECT	Radiomics	HMLSs (LOLIMOT, RBF, RNN, MLP-B)	Hold-out	–	–	–	–	(Mild, Moderate, Severe) UPDRS III: (22.82, 24.22, 25.41); MMSE: (31.75, 32.6, 33.72)
133	Bikias et al. (2021)	2021	11 PD	Gait	–	CNN	LOOCV and 10-fold	–	83, 88	86, 90	–	–
134	Masud et al. (2021)	2021	564 PD vs 192 CN	Voice	–	SAE	K-fold	96.4	97.1	98.5	–	–
135	Vyas et al. (2021)	2021	236 PD vs 82 CN	MRI	–	2D CNN 3D CNN	Hold-out	88.9 72.22	92.0 100	–	86.2 50.0	F1-score: 92.3 F1-score: 83.7
136	Solana-Lavalle and Rosas-Romero (2021)	2021	Male: 226 PD vs 86 CN Female: 104 PD vs 64 CN	MRI	FSOS, PCA	KNN, MLP, SVM, RF, NB, LR, BN	10-fold	99.01 96.97	99.35 100	100 96.15	–	–

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Table 1 (*continued*).

Sr. No	Study	Year	Subjects	Modality	Feature extraction	Architecture	Cross validation	Performance (in %)				
								Acc	Sen	Spec	AUC	MISC
137	Cantürk (2021a)	2021	25 PD vs 15 CN	Hand-drawn	AlexNet, GoogleNet	KNN, SVM, AlexNet, GoogleNet	10-fold	94.0	93.0	95.0	–	95.0 F1-score
138	Bhan, Kapoor, Gulati, and Goyal (2021)	2021	30 PD vs 24 CN	MRI	–	LeNet-5	Hold-out	97.92	98.0	–	–	98.0 (F1-score)
139	Balaji, Brindha, Elumalai, and Vikrama (2021)	2021	93 PD vs 73 CN	MRI	FSOS, PCA	CNN–LST (Multi-class) CNN–LST (Binary class)	10-fold	96.60 98.60	96.20 98.23	98.08 99.10	–	–
140	Loh, Ooi, et al. (2021)	2021	15 PD vs 16 CN	EEG	GT	2D-CNN	10-fold	99.46	99.48	–	–	99.46 (F1-score)
141	Yadav (2021)	2021	15 PD vs 15 CN	fMRI	–	Bayesian 3D-CNN	Hold-out	97.92	–	–	–	–
142	Talai, Sedlacik, Boelmans, and Forkert (2021)	2021	45 PD, 20 RPS-RS, 38 CN	MRI	–	SVM + MLP	Hold-out	95.1	–	100	–	100 (F1-score)
143	Heidariv-incheh et al. (2021)	2021	5 PD vs 5 CN	MRI	–	MCPD-Net	Hold-out	–	77.0	–	–	66.0 (F1-score)

MISC - Miscellaneous, **Acc** - Accuracy, **Sen** - Sensitivity, **Spec** - Specificity, **MSE** - Mean squared error, **AUC** - Area under curve, **CC** - correlation coefficients, **PCC** - Pearson s correlation coefficients, **STFT** - Short-time Fourier transformation, **NMF** - Non-negative matrix factorization, **LSTM** - Long short-term memory, **BLSTM** - Bidirectional long short-term Memory, **PD** - Parkinson's disease, **CN** - Controlled normal/healthy control, **RPS-RS** - Progressive supranuclear palsy Richardson's syndrome, **DNN** - Deep neural network, **ELM** - Extreme learning machine, **CNN** - Convolutional neural network, **LOOCV** - Leave one out Cross-validation, **MRI** - Magnetic resonance imaging, **CI** - Clinical Information/Interpretability, **CF** - Clinical Features, **BGRU** - Bidirectional gated recurrent unit, **SVM** - Support vector machine, **ECOC-SVM** - Error-correcting output code (ECOC) using SVM **RVSM** - Relaxed variable splitting method, **BoS** - Bag of samples, **GRU** - Gated recurrent unit, **HWI** - Handwriting images, **HWS** - Handwriting samples, **MLP** - Multilayer perceptron, **EEG** - Electroencephalography, **SPECT** - Single photon emission computed tomography, **VEOG** - Vertical electrooculography, **STFT** - Short time Fourier transform, **PET** - Positron emission tomography, **CAE** - Convolutional autoencoder, **SAE** - Stacked autoencoder, **KNN** - K-nearest neighbour, **IGKNN** - Information Gain Algorithm-based KNN Hybrid Model **RBM** - Restricted Boltzmann machine, **RP** - Recurrence plot, **SOM** - Self-organizing maps, **NIPALS** - Nonlinear iterative partial least squares, **ISVR** - Immune support vector regression, **MAE** - Mean absolute error, **UPDRS** - Unified Parkinson's disease rating scale, **ImgSpect** - Image spectrogram, **PSR** - Pupil size variations, **LFP** - Local field potential, **DWT** - Discrete wavelet transform, **TQWT** - Tunable Q-factor Wavelet Transform, **GT** - Gabor transform, **HFM** - Hand and finger movement, **RNN** - Recurrent neural network, **HMV** - Hand movement video, **ANN** - Artificial neural networks, **EANNs** - Evolutionary artificial neural networks, **VGRF** - Vertical ground force signals, **DaT Scan** - Dopamine transporter scan, **DaT SPECT** - Dopamine transporter SPECT **DT** - Decision tree, **GBDT** - Gradient boosting decision tree, **NB** - Naive Bayes, **RF** - Random forest, **HMLSs** - Hybrid machine learning systems, **GA** - Genetic algorithm, **QSMI** - Quantitative susceptibility mapping images, **AE** - Autoencoder, **ROC** - Receiver operating characteristics, **Sp-AE** - Sparse autoencoder, **FFT** - Fast Fourier transform, **KCFW** - K-means clustering feature weighting method, **CVANN** - Complex valued artificial neural network, **LMD** - Large margin distribution, **ESN** - Echo state network, **RLSFN** - Random least squares feed forward neural network, **WSS** - Wearable sensor signals, **OPF** - Optimum path forest, **ENN** - Evolutionary neural network, **TD** - Tensor decomposition, **FDG-PET** - Fluorodeoxyglucose positron emission tomography, **CMCRVFL** - Cascaded multi-column random vector functional link, **Q-BTDNN** - Q- backpropagated time delay neural network, **WNN** - Wavelet neural network, **DBN** - Deep belief network, **FCN** - Fully connected network, **MP-NN** - Multilayer perceptron neural network, **MLP-ANN** - Multilayer perceptron artificial neural network, **PNN** - Probabilistic neural network, *GRNN* - general regression neural network, **BPNN** - backpropagation neural network, **mRMR** - Minimum redundancy maximum relevance (mRMR), **DRA** - Dimensionality reduction algorithms, **HMM Net** - Hidden Markov model network, **CAPSO-MLP** - Centripetal accelerated particle swarm optimization multilayer perceptron **PSO-MLP** - Particle swarm optimization multilayer perceptron, **ICA-MLP** - Imperialist competitive algorithm multilayer perceptron, **GSA-MLP** - Gravitational search algorithm multilayer perceptron, **SFS** - Sequential forward selection, **SFFS** - Sequential forward floating search, **LR** - Logistic regression, **BayNet** - Bayesian network, **PCA** - Principal component analysis, **LDA** - Linear-discriminant analysis **ICA** - Independent component analysis, **HOS** - Higher-order statistics, **LOF** - Lower-order features, *FSOS* - First- and second-order statistics, **Acclrm** - Accelerometer, **EMG** - Electromyography, **PBL-McRBFN** - Projection based learning meta-cognitive neural classifier approach, **fMRI** - Functional Magnetic Resonance Imaging.

Table 2
Comparison of model architecture.

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
1	Ghare-hchopogh and Mohammadi (2013)	Classif.	3	1 (22)	1 (5)	1 (1)	tangent sigmoid	–	–	–	–	–	–	–	–	–	–	–
2	Hossen (2013)	Classif.	3	1 (16)	1 (3)	1 (1)	hyperbolic tangent sigmoid	–	–	–	–	–	–	–	–	–	–	–
3	Babu and Suresh (2013)	Classif.	3	1 (–)	1 (–)	1 (–)	Gaussian	–	–	–	–	–	–	–	–	4 lears. strat.	–	–
4	Mandal and Sairam (2013)	Classif.	1	–(–)	1 (–)	–(–)	–	–	–	–	–	–	RMS	–	–	Dynamic	–	–
5	Hosseini and Makki (2013)	Classif.	5	1 (48)	3 (25, 1, 25)	1 (48)	sigmoid	–	–	–	–	–	MSE	–	–	0.003,	$\beta = 0.5$	–
6	Hariharan et al. (2014)	Classif.	4	1 (–)	2 (–)	1 (–)										0.01 to 0.1		
7	Khorasani and Daliri (2014)	Classif.	2	–(–)	1 (–)	–(–)	–	–	–	–	–	–	–	–	–	–	–	–
8	Sharma and Giri (2014)	Classif.	3	1 (–)	1 (–)	–(–)	RBF	–	–	–	–	–	MSE	–	–	–	–	
9	Beheshti et al. (2014)	Classif.	3	1 (–)	1 (–)	1 (–)	RBF	–	–	–	–	–	MSE	–	–	–	–	

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
10	Islam et al. (2014)	Classif.	3	1 (–)	1 (–)	1 (–)	–	–	–	–	–	–						
11	Olanrewaju et al. (2014)	Classif.	3	1 (8)	1 (10)	1	Logsigmoid	–	–	–	–	–	–	–	–	–	–	–
12	Joo et al. (2014)	Classif.	3	1 (–)	1 (–)	1 (–)	Sigmoid	–	–	–	–	–	–	–	–	–	–	–
13	Shaikh (2014)	Classif.	3	1 (–)	1 (–)	1 (–)	–	–	–	–	–	–	RMS		500	–	–	–
14	Mandal and Sairam (2014)	Classif.	–(–)	–(–)	–(–)	–	sigmoid	–	–	–	–	–	–		–	Dynamic	–	–
15	Froelich et al. (2014)	Classif.	1	1 (50)	–(–)	–(–)	–	–	–	–	–	–	–	–	100	–	–	–
16	Peker et al. (2015)	Classif.	3	1 (–)	1 (–)	1 (–)	Complex Sigmoid	–	–	–	Range [0, 1]		–	–		>0		
17	Hirschauer et al. (2015)	Classif.	4	1 (8)	2 (–)	1 (–)	–	–	–	–		–	–		100 iterations			
18	Saloni et al. (2015)	Classif.	–	–(–)	–(–)	–(–)	Tan sigmoid, Logsigmoid, Linear	–	–	–	–	–	MSE	–				
19	Er et al. (2016)	Classif.	4	1 (–)	2 (–)	1 (–)	sigmoid	–					RMS					

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
20	Hlavica et al. (2016)	Classif. (ANN, ANFIS)	3	1 (–)	1 (–)	1 (–)	–	–	–	–	Range [0, 1]		MSE	GD	–	–	–	–
			5	1 (–)	3 (–)	1 (–)												
21	Eskofier et al. (2016)	Classif.	6	1 (–)	4 (–)	1 (–)	ReLU	2 FC layer					CE	Adam		0.001		
22	Jane et al. (2016)	Classif.	4	1 (–)	2 (–)	1 (–)	Sigmoid	–	–	–	–	–			100, 150, 200	0.01		
23	Khan et al. (2016)	Classif.	3	1 (–)	1 (–)	1 (–)	–	–	–	–	–	–	–	–	50	0.01	–	–
24	Al-Fatlawi et al. (2016)	Classif.	3	1 (–)	2 (–)	1 (–)	Logistic	–	–	–	–	–	MSE	–	–	0.001	–	
25	Pereira et al. (2016)	Classif.	5	–(–)	–(–)	–(–)	ReLU	–	–	–	–	–	Softmax	–	25	[0, 0.01]	–	LR: ϵ [0, 0.01], β ϵ [0, 1], WD: ϵ[0, 0.001]
26	Turner et al. (2016)	Classif.	3	1 (–)	1 (–)	1 (–)	Sigmoid	–	–	–	–	–	–	–	–	–	–	–
27	Abiyev and Abizade (2016)	Classif.	6	1 (–)	4 (–)	1 (–)	(–)	–	–	–	–	–	RMSE		2000	[0-1]	–	–
28	Sivanesan et al. (2016)	Classif.	4	1 (–)	2 (–)	1 (–)	–	–	–	–	–	–	MSE	–	5000, 235, 144, 215	–		–
29	Gürüler (2017)	Classif.	3	1 FC layer (–)	–	1 Output layer (–)	Sigmoid	–	–	–	SE	–					–	–
30	Zia et al. (2016)	Pred.	3	1 (–)	1 (–)	1 (–)	–				–	–	–	–	48 different trials			48 different trials
31	Challa et al. (2016)	Pred	10	1 (–)	8 (–)	1 (–)	–	–	–	–	–	–	–	–	–	0.4	–	–
32	Tagaris et al. (2017)	Classif.	–	ResNet50	FC layer (128, 128), BLSTM (128)	–	ReLU	–	–	Dropout (0.5)	–	–	MSE	Adam	12000	0.001	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
33	Choi et al. (2017)	Classif.	6	2 (7 ×7 ×7 ×16 (stride = 4) + MaxPool 3 ×3 ×3 (stride = 2))	1 (5 ×5 ×5 ×16 (stride = 1) + MaxPool 3 ×3 ×3 (stride = 2))	1 ((3 ×3 ×3 ×64 (stride = 1)), FC layer (256, 2))	ReLU, Softmax	2	–	–	–	–	–	SGDM	30	1×10^{-4}, $\beta = 0.9$	Log decrease to 1 $\times 10^{-6}$ at final epoch	–
34	Zhang (2017)	Classif.	–	2 (9, 8)	–	–	–	2	–	L2 (0.5)	–	–	–	–	–	–	–	GS
35	Pereira et al. (2017)	Classif.	2	–(–)	1 (–)	–(–)	–	2	–	–	–	–	–	CDA	1000	0.1	–	GS
36	Nilashi et al. (2018)	Classif.	1	SOM (3 ×3)	–	–	–	–	–	–	–	–	–	–	–	–	–	–
37	Gómez-Vilda et al. (2017)	Classif.	1		–	1 RVFL Layer (–)	–	–	–	–	–	–	–	–	200	–	–	–
38	Nikandish and Kheirkhah (2017)	Classif.	3	1 FC layer (30)	–	1 Output Layer (7)	–	–	–	–	–	–	–	GD	2000	–	–	–
39	Um et al. (2017)	Classif.	10	1 (4 ×1 ×16)	6 (4 ×1 ×32, 3 ×1 ×64, 3 ×3 ×64, 2 ×3 ×64, 2 ×3 ×64, 2 ×3 ×64) + GAP													
	FC layer (48, 2)	ReLU, Softmax	–	–	–	BN	before ReLU	–	–	400	–	–	–					
40	Zhang et al. (2018)	Classif.	4	10 ×10 ×64 (stride = 2)	10 ×10 (stride = 2) + MaxPool 2 ×2 (stride = 2)	–	ReLU, Softmax	–	–	–	–	–	–	–	–	–	–	–
41	Passos et al. (2018)	Classif.	–	–	Resnet50	–	–	–	–	–	–	–	–	–	–	–	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
42	Shi et al. (2018)	–	–	max 5 HL, max 9 columns	–	–	–	–	–	–	–	–	–	–	–	–	–	–
43	Zhao et al. (2018)	Classif.	6 (Multi-path)	5 ×5 + MaxPool LSTM (–)	5 ×5 + MaxPool, Flatten, Softmax LSTM (–), Softmax	FC layer (4 or 2)	–	3	–	Dropout (0.5)	–	–	–	–	300000 and 500000	0.001	–	–
44	Khojasteh et al. (2018)	Classif.	8	1 (3^2 ×32 + maxpool 2^2)	2 (3 ×3 ×3 ×16 + maxpool 2^2, 3 ×3 ×3 ×16 + MaxPool 2^2)	2 FC layer (512, 2)	ReLU, Softmax	2	–	–	–	–	–	SGDM	100	0.01, $\beta = 0.9$	–	–
45	Kim, Wit, and Thurston (2018)	Classif.	–	Inception v3	–	–	–	–	–	–	–	–	–	–	20000	0.01	–	–
46	Rumman et al. (2018)	Classif.	–	–	–	–	Sigmoid	–	–	–	–	–	–	–	–	–	–	–
47	Alaskar and Hussain (2018)	Classif.	3	1 FC layer (20)	–	1 Output Layer (1)	–	–	–	–	–	–	–	LM	2000	–	–	–
48	Adams et al. (2018)	Classif.	9	1 (7 ×7 ×7 ×16 (stride = 3) + Pool 3 ×3 ×3 (stride = 2))	3 (5 ×5 ×5 ×16 (stride= 1) + Pool 3 ×3 ×3 (stride = 2), 3 ×3 ×3 ×64 (stride = 1) + Pool 3 ×3 ×3 (stride = 2), Flatten)	FC layer (1024, 2)	ReLU, Softmax	–	–	Dropout (0.5)	–	–	CE	Adam	50	0.001	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
49	Khatamino et al. (2018)	Classif.	6	1 (3 ×3 (stride = 1)) + MaxPool 2 (stride = 2)	1 (3 ×3 (stride = 1)) + MaxPool 2 (stride = 2)	FC layer (128, 2)	–	–	–	Dropout (0.5)	–	–	–	SGD, Adam, Adadelta, RMSprop	100	0.01	–	–
50	Wu et al. (2018)	Classif.	–	1 (Conv (–) + Pool) (–)	3 (Conv (–) + Pool (–), Conv (–) + Pool (–), Conv (–) + Pool (–))	FC layer (–)	–	–	–	–	–	–	–	–	–	–	–	–
51	Wroge et al. (2018)	Classif.	6	3 HL + BN + Dropout	–	–	Sigmoid	–	–	–	–	–	–	–	–	–	–	–
52	Gallicchio et al. (2018)	Classif.	–	Recurrent units per layer: (10, 20, 30, 40, 50)	–	–	tanh	–	–	RR: (0, 10^{-10}, 10^{-9}, …, 10^{0}), SR: (0.7, 0.8, 0.9, 1.0)	Input scaling: (0.1, 0.5, 1, 2), Interlayer scaling: (0.1, 0.5, 1, 2)	–	–	–	–	–	–	–
53	Chen et al. (2018)	Classif.	–	4 PN Layers (–)	–	–	–	–	–	–	–	–	–	–	–	–	–	–
54	Esmaeilzadeh et al. (2018)	Classif.	12	1 (3 ×3 ×3 ×32 + MaxPool)	5 (3 ×3 ×3 ×32 + MaxPool, 3 ×3 ×3 ×64 + MaxPool, 3 ×3 ×3 ×64 + MaxPool 3 ×3 ×3 ×128 + MaxPool, 3 ×3 ×3 ×128 + MaxPool)	3 (512, 128)	LReLU	–	Bias (0.05), Kernel (0.001)	Dropout	Group, Batch	–	CE	Adam	147	0.00005	Exp. LR decay	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
55	Pereira et al. (2018)	Classif.	10	1 (11 ×11 ×– (stride = 4) + Pool)	4 (5 ×5 ×-, 3 ×3 ×–, two inner product layers)	1 (–)	ReLU, Softmax	–	–	Dropout	–	–	CE	–	10000	–	–	–
56	Oh et al. (2018)	Classif.	13	1 (20 ×1 ×5 + MaxPool)	3 (10 ×1 ×10 + MaxPool, 10 ×1 ×10 + MaxPool, 5 ×1 ×5 + MaxPool)	2 FC layer (20, 10), Output layer (2)	ReLU, Softmax	–	–	Dropout (0.5)	–	–	CE	Adam	1588	0.0001	–	–
57	Vásquez-Correa et al. (2018)	Classif.	–	Filter size: (3, 5, 7), Depth (4, 8, 16, 32, 64),	FCHU: (16, 32, 64, 128)	–	–	–	–	Dropout [0.1, 0.9]	–	–	CE	SGD	150	0.0001, 0.0005, 0.01	Exp. LR decay	BayOpt.
58	Hwang et al. (2018)	ImgRect., denoising	22	3 (3 ×3 ×2 + MaxPool, 3 ×3 ×2 + MaxPool, 3 ×3 ×2 + MaxPool)	17 (3 blocks of (3 ×3 ×–+ MaxPool), Deconv, 2 blocks of (3 ×3 ×–), Deconv)	2 (3 ×3 ×–, 3 ×3 ×–)	ReLU	0	Dropout (0.7)	–	–	–	RMSE	Adam	6	–	–	–
59	Grover et al. (2018)	Classif.	4	1 (10)	2 (20, 10)	1 (2)	Softmax	4	–	4	min–max	At input layer	–	GD	–	–	–	–
60	(Vásquez-Correa et al., 2018)	Classif.	–	Filter size: (3, 5, 7), Depth: (4, 8, 16, 32, 64),	FCHU: (16, 32, 64, 128)	–	–	–	–	Dropout [0.1, 0.9]						0.0001, 0.0005, 0.01		BayOpt.

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Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
61	Kim, Lee, et al. (2018)	Classif.	4	1 (3 ×50 ×30 (stride = 3 ×10))	1 (1 ×20 ×30 (stride = 1))	FC layer (2310, 4)	ReLU, Softmax		Dropout (0.3) (after ReLU)	–	–	–	CE	Adam	200	0.001	–	GS
62	Gong et al. (2018)	Classif.	3	(1, 2, 3) HL	–	–	Sigmoid	–	–	–	–	–	–	SubGD	–	–	–	GS
63	Lacy et al. (2018)	Classif.	–	–	–	–	LF	–	–	–	–	–	–	–	–	–	–	–
64	Parisi et al. (2018)	Classif.	3	–	1 HL (3)	–	Sigmoid	–	–	–	–	–	MSE	GD	67		–	–
65	Khan et al. (2018)	Classif.	3	1 FC layer (–)	–	1 Output layer (–)	–	–	–	–	–	–	–	–	50	–	–	–
66	Martinez-Murcia et al. (2017)	Classif.	6	1 (5 ×5 ×5 ×18 + MaxPool 2 ×2 ×2	1 (5 ×5 ×5 ×1 ×16 + MaxPool 2 ×2 ×2	FC layer (4000)	ReLU	–	–	Dropout (0.5)	–	–	–	–	–	–	–	–
67	Wodzinski et al. (2019)	Classif.	–	CAE	–	FC layer	–	–	–	–	–	–	–	–	–	–	–	–
68	Xiao et al. (2019)	Classif.	–	Conv + BN + ReLU	(DB, (Conv + BN + ReLU + Pool), (Conv + BN + ReLU + Pool))	FC layer, Output Layer	ReLU, Sigmoid	–	–	–	–	–	–	–	–	–	–	–
69	Peng et al. (2019)	Classif.	–	AE	–	–	–	–	–	–	–	–	–	–	–	–	–	
70	Qin et al. (2019)	Classif.	5	1 ×1 ×256	1 ×1 ×64, 1 ×1 ×256, 1 ×1 ×64	FC (512, 2)	Softmax	–	L2 (10^{-5})	L2 (10^{-7})	Norm	Input feature	CE	–	680K	10^{-4}	–	–
71	Masiala et al. (2019)	Classif.	2	LSTM (–)	LSTM (–)	FC layer (–, –)	–	–	–	–	–	–	–	–	–	–	–	–
72	Tang et al. (2019)	Classif.	1	LSTM (–)	–	–	–	–	–	–	–	–	–	–	–	–	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
73	Kadam and Jadhav (2019)	Classif.	3 (multi path)	AE (–)	AE (–)	Softmax Reg. layer 2			KL (0.5 to 1), L2 (0.003 to 0.9)		Softmax	MSE	SCG	100	–	–	–	
				–	Softmax Reg. layer 1	–	–	–	–	–								–
					Concat layer	Softmax Reg. layer 3	–	–	–	–	–	–	–	–	–	–		
74	Mikos et al. (2019)	Classif.	3	1 FC layer (5)	–	1 Output layer (1)	–	–	–	–	–	–	CE	GD	–	–	–	–
75	Afonso et al. (2019)	Classif.	–	1 (11 ×11 ×– (stride = 4) + Pool)	4 (5 ×5 ×-, 3 ×3 ×–, two inner product layers)	1 (–)	ReLU, Softmax	–	–	–	–	–	–	–	–	–	–	–
76	Ribeiro et al. (2019)	Classif.	2	2 (64)	–	GRU	ReLU	–	–	Dropout (0.5)	–	–	CE	Adam	30	5×10^{-4} with norm clipping (1)	–	GE
77	Johri and Tripathi (2019)	Classif.	6	1 (3^2 ×32 (stride = 1) + MaxPool 2 ×2 (stride = 2))	1 (3^2 ×16 (stride = 1) + MaxPool 2 ×2 (stride = 2))	FC layer (16, 1)	–	–	Dropout (0.5)	–	–	–	CE	Adam	25	–	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
78	Naseer et al. (2020)	Classif.	8	1 (11 ×11 ×96 (stride= 4))	4 (5 ×5 ×256, 3 ×3 ×384, 3 ×3 ×384, 3 ×3 ×256)	3 (4096, 4096, 1)	–	–	–	L2 (0.0005)	–	–	CE	SGD + momentum ($\beta = 0.9$)	30	0.0001	–	–
79	Xue and Xin (2019)	Classif.	8	1 (32 ×50 ×50 + MaxPool)	2 (32 ×25 ×25 + MaxPool, 32 ×13 ×13 + MaxPool)	2 (128, 2)	–	2	–	L0 with RVSM	–	–	CE	SGD	20	–	–	–
80	Ortiz et al. (2019)	Classif.	12	1 (7 ×7 ×7 ×41 + MaxPool 2 ×2 ×2)	4 (5 ×5 ×5 ×21 + MaxPool 2 ×2 ×2, 5 ×5 ×5 ×11, 4^3 ×11, 3 ×3 ×3 ×11 + MaxPool 2 ×2 ×2)	FC layer (2048, 2048, 2)	ReLU, Softmax	3	–	Dropout (0.7)	–	–	CE	–	–	–	–	–
81	Vital et al. (2020)	Classif.	–	–	–	1 (10)	1 (1)	–	–	–	–	–	–	–	1000	–	–	–
82	Gil-Martín et al. (2019)	Classif.	7	1 ×5 ×16 + MaxPool 1 ×3	1 ×5 ×16 + Flatten	FC layer (128, 32, 1)	–	–	–	Dropout (0.2)	–	ReLU, Sigmoid		RMSprop	25	–	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
83	Kollia et al. (2019)	Classif.	52	Resnet50 + GRU	–	–	ReLU, Softmax	–	–	–	–	–	–	GPA	–	–	–	–
84	Ali et al. (2019)	Classif.	3	1 FC layer (–)	1 FC layer (–)	1 Output layer (–)	–	–	–	–	–	–	CE	–	–	–	–	–
85	Mostafa et al. (2019)	Classif.	–	–	–	–	–	–	–	–	–	–	MAE	–	–	–	–	–
86	Sivaranjini and Sujatha (2019)	Classif.	8	11 ×11 (stride = 4) + Act + Norm + Pool	5 ×5 (stride = 1) + Act + Norm + Pool, 3 ×3 (stride = 1) + Act, 3 ×3 (stride = 1) + Act, 3 ×3 (stride = 1) + Act + Pool	–	ReLU, Softmax	–	–	Dropout (0.5)	–	Softmax	CE	SGDM	30	0.0001, β = 0.9	–	–
87	Golshan et al. (2020)	Classif.	20	3 (3 ×24 (stride = 1) + BN),	7 (3 ×3 ×3 ×48 (stride = 1) + MaxPool + BN, 3 ×3 ×3 ×48 (stride = 1) + BN, 3 ×3 ×3 ×24 (stride = 1) + Maxpool + BN, 3 ×3 ×3 ×48 (stride = 1) + MaxPool + BN)	FC layer (64, 3)	ReLU, Softmax	2	–	Dropout	BN	After ReLU	CE	Adam	30	–	–	–

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Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
88	Pang et al. (2020)	Classif.	–	Pretrained Oldpose model	–	–	–	–	–	L2 (0.0001)	–	–	–	SGDM	–	0.0001, $\beta = 0.9$		
89	Alfaro-Ponce and Chairez (2020)	Classif.	–	–	–	–	–	–	–	—-	–	–	–	–	–	–	–	–
90	Lin et al. (2020)	Classif.	12	2 (LSTM (–) + Dropout)	6 (LSTM (–) + Repeat vector, LSTM (–) + Dropout, LSTM (–) + Concat layer)	FC layer (64, 32 , 16, 1)	Sigmoid, tanh	–	–	L2 (0.1, 0.2, …, 0.5), Dropout (0.1, 0.2, …, 0.5)	–	–	MSE	Adam	–	–	–	–
91	El Maachi et al. (2020)	Classif.	8	1 (3 ×1 ×8 + MaxPool 2 ×1)	3 (3 ×1 ×16 + MaxPool 2 ×1, 3 ×1 ×16 + MaxPool 2 ×1, 3 ×1 ×16 + MaxPool 2 ×1)	FC layer (100), Concat layer, FC layer (100, 20, 1)	SELU, Sigmoid	3	–	Dropout (0.5)	–	–	–	NADAM	–	0.001	halving LR	–
92	Szumilas et al. (2020)	Classif.	–	–	1 (50)	1 (1)	–	–	–	–	–	–	–	–	–	–	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
93	Kaur et al. (2020a)	Classif.	Disc - 5	1 (3 ×3 ×32 (stride = 2) + LReLU)	1 (3 ×3 ×64 (stride = 2) + LReLU), 1 (3 ×3 ×128 (stride = 2) + LReLU), 1 (3 ×3 ×256 (stride = 2) + LReLU), Flatten	FC layer (2)	LReLU, Softmax	1	–	–	–	–	–	SGD	–	–	–	
			Gen - 6	1 FC Layer (5120)	1 Dconv (3 ×3 ×256 (stride = 2) + ReLU), 1 Dconv (3 ×3 ×128 (stride = 2) + ReLU), 1 Dconv (3 ×3 ×64 (stride = 2) + ReLU)	1 Dconv (3 ×3 ×32 (stride = 2) + Tanh)	ReLU, TanH	–	–	–	–	–	–	SGD	–	–	–	
			Clf - 12	AlexNet (pretrained)	1 FC layer (4096)	1 FC layer (2)	ReLU, Softmax	2	–	Weight decay (0.06)	–	–	–	Adam	200	0.001, β = 0.9	–	GS
94	Li et al. (2020)	Classif.	6	1 (1 ×64) + BN + ReLU + MaxPool + SE)	1 (1 ×64 + BN + ReLU + MaxPool + SE), 1 (1 ×64 + BN + ReLU)), 1 LSTM (128) + Attention	1 FC (1)	ReLU, Sigmoid	1	–	–	BN	Before activation	–	–	–	0.001	Decrease by 8	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
95	Wingate et al. (2020)	Classif.	52	ResNet-50	GRU (128)	Dense (1)	ReLU, Sigmoid	2	–	Dropout	–	–	–	–	–	0.001	–	–
96	Wang et al. (2020)	Classif.	3	Input (13)	FC (20)	Dense (1)	Sigmoid	2	–	Dropout	BN	–	CE	SGD	100	–	–	–
97	Kaur et al. (2020b)	Classif.	3	Input (22)	[2–13] (1, 2, ..40)	Dense (1)	(ReLU, Sigmoid, TanH, Linear)	[2–14]	–	–	BN	–	CE	SGD	100	–	–	GSO
98	Choi et al. (2020)	Classif.	10	1 (5 ×5 ×5 (stride =4))	1 (5 ×5 ×5 (stride =2)), 2 (5 ×5 ×5) + GAP + 1 Flatten (128)	1 (1)	ReLU, Sigmoid	2	–	–	–	–	CE	Adam	–	–	–	–
99	Zhang et al. (2020)	Classif.	8	1 (1 ×8 + BN + MaxPool)	1 (1 ×16 + BN + MaxPool), 2 (1 ×32 + BN + MaxPool), 2 (1 ×64 + BN + MaxPool), 3 (1 ×128 + BN + MaxPool)	1 Fc (1)	Sigmoid (FC)	1	–	–	–	–	–	–	100	–	–	–
100	Ashour et al. (2020)	Classif.	3	1 (100)	–	1 (2)	–	2	–	–	–	–	–	SGD	–	–	–	–
101	Akyol (2020)	Classif.	6	1 (2)	4 (4, 8, 16, 32)	1 (64)	ReLU	6	–	–	–	–	–	SGD	100	–	–	–
102	Olivares et al. (2020b)	Classif.	3	1 (23)	1 (12)	1 (1)	Sigmoid	2	–	–	Range [0, 1]	At input layer	–	–	1	–	–	–
103	Sigcha et al. (2020)	Classif.	9	1 (4 ×1 ×128 + MaxPool)	1 (4 ×1 ×128 + MaxPool), 1 Flatten (896), 1 LSTM (64)	3 (80, 40, 1)	ReLU	3	–	–	–	–	CE	Adam	250	1×10^{-4}	–	RS

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Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
104	Bouza et al. (2020)	Classif.	5	2 (–)	3 (–)	1 (–)	TReLU	2	–	–	–	–	CE	Adam	200	0.005	–	–
105	Oktay and Kocer (2020)	Classif.	5	1 (1D Conv (20 ×30) + MaxPool)	1 (1D Conv (10 ×30) + MaxPool)	LSTM (–)	ReLU, Sigmoid	1	–	–	–	–	CE	SGD	–	–	–	–
106	Shahtalebi et al. (2020)	Reg.	4	2 BGRU (4)	–	2 BGRU (4)	ReLU	–	–	–	–	–	MSE	Adam		0.0001	–	GS
107	Pianpanit et al. (2020)	Classif.	6	2 (7 ×7 ×7 ×16 (stride = 4) + MaxPool 3 ×3 ×3 (stride = 2))	1 (5 ×5 ×5 ×16 (stride = 1) + MaxPool 3 ×3 ×3 (stride = 2))	1 ((3 ×3 ×3 ×64 (stride = 1)), FC layer (256, 2))	ReLU, Softmax	2	–	–	–	–	–	SGDM	30	1×10^{-4}, $\beta = 0.9$	log decrease to 1×10^{-6} at final epoch	–
			11	1 ((5 ×5 ×5 ×16 (stride = 1) + MaxPool 3 ×3 ×3 (stride = 2))	8 ((3 ×3 ×3 ×32 (stride = 1) + MaxPool 3 ×3 ×3 (stride = 2), (3 ×3 ×3 ×64 (stride = 1) + Maxpool 3 ×3 ×3 (stride = 2), (2 ×3 ×2 ×256 (stride = 1))	FC layer (256)	ReLU, sigmoid	1	–	–	BN	After ReLU	–	SGDM	30	1×10^{-4} $\beta = 0.9$	log decrease to 10^{-6} at final epoch	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
108	Zanca et al. (2020)	Classif.	6	5 ×1 ×16 + Pool 2 ×1	3 ×1 ×32 + Flatten	FC layer (10, 2)	Softmax	2	Dropout	–	BN	After input	–	–	–	–	–	–
109	Rasheed et al. (2020)	Classif.	3	1 (–)	1 (–)	1 (–)	–	–	–	–	–	–	–		500	–	Exp till	–
110	Wanjale et al. (2020)	Classif.	5	1 (16)	3 (100, 200, 100)	1 (2)	ReLU	–	–	–	–	–	LR	Adam	500	1.46×10^{-3}	–	–
111	Chakraborty et al. (2020)	Classif.	7	1 (20)	5 (10, 128, 256, 64, 16)	1 (3)	–	–	–	–	–	–	CE	AdaDelta	–	1	0.8	–
112	Veeraragavan et al. (2020)	Classif.	3	1 (34)	1	1 (2)	–	–	–	–	min–max	–	–	–	–	–		–
113	Olivares et al. (2020b)	Classif.	3	1 (23)	1 (12)	1 (1)	Sigmoid	2	–	–	Range [0, 1]	At input layer	–	–	1	–	–	–
114	Huang et al. (2020)	Classif.	–	–	–	–	ReLU	–	–	–		–	–	RMS prop Adam	–	–		–
115	Kamran et al. (2021)	Classif.	12	AlexNet (Pre-trained)	–	–	ReLU	–	–	L2 (0.0005)		–	–	SGD	0.0001 , $\beta = 0.7$	–	–	–
116	Chien et al. (2021a)	Classif.	9	1 (–)	5 (–)	3 (–)	–	3	–	–	–	–	–	–	–	–	–	–
117	Mohammed et al. (2020)	Classif.	12	1 (3 ×3 + ReLU + MaxPool)	2 (3 ×3 + ReLU + MaxPool), 1 Dense (–)	1 Dense (1)	ReLU, Sigmoid	2	Dropout	Dropout	–	–	CE	Adam	–	10^{-3}	Exp till 10^{-8}	–
118	Yasaka et al. (2021)	Classif.	8	1 (1 ×1 ×8 + BN + MaxPool)	1 (1 ×16 + BN + MaxPool), 2 (1 ×32 + BN + MaxPool), 2 (1 ×64 + BN + MaxPool),	1 Fc (2)	ReLU, Softmax	3	–	–	–	–	CE	AdaGrad	20	–	–	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
119	Quan et al. (2021)	Classif.	3	–	BLSTM (20, 50, 100, 200)	Dense (1)	(ReLU, Sigmoid, TanH, Linear)	2	–	Dropout, R-Dropout	–	–	–	(Adam, Nadam , Adagrad, SGD)	10, 30, 50, 100, 150	(1, 0.1, 1e−2, 1e-4)	–	GS
120	Adams et al. (2021)	Classif.	6	1 ((7 ×7 ×7 ×16 (stride = 4) + MaxPool 3 ×3 ×3 (stride = 2))	1 (5 ×5 ×5 ×16 (stride = 1) + MaxPool 3 ×3 ×3 (stride = 2)); 1 (3 ×3 ×3 ×32 (stride = 2)) + 1 (3 ×3 ×3 (stride = 1) + Maxpool 3 ×3 ×3 (stride = 2))	3 FC (256, 64, 16)	ReLU	3	–	Dropout (0.5)	0–1	At input	CE	Adam	50	0.001	–	–
121	Demir et al. (2021)	Classif.	7	SeqFold + 1(5 ×5x3 ×20) + BN	SeqUnfold + Flatten + LSTM (100)	2 FC(350, 2)	ReLU, Softmax	2	–	Dropout (0.4)	Feature	–	CE	SGDM	200	0.001	–	–
122	Pahuja and Nagabhushan (2021)	Classif.	3	Input	1 FC(10)	1 FC(2)	–	2	–	–	–	–	–	LM, CGD	–	–	–	–
123	Varrecchia et al. (2021)	Classif.	3	1 FC(–)	2/3 FC(20-50)	1 FC(2)	–	–	2–6	–	–	–	MSE	LM	1000	–		GS
124	Shichkina et al. (2021)	Classif.	5	2 LSTM (128)	2 FC(64, 32)	1 FC(1)	ReLU Sigmoid	3	–	–	–	–	CE, MSE	–	90	–	–	GA
125	Bikias et al. (2021)	Classif.	–	1 (10 ×100 + MaxPool 3 ×1)	1(10 ×40 + GAP 2 ×1) + Flatten + Dropout (0.5)	2 FC()	ReLU, Softmax	2	–	Dropout (0.5), Early stopping	–	–	CE	Adam	–	–	–	GS

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
126	Bikias et al. (2021)	Classif.	4	1 FC(754)	2 FC(3, 4)	1 FC (2)	Sigmoid, Softmax	4	–	–	–	–	CE	GD	–	–	–	–
127	Vyas et al. (2021)	Classif.	–	2 (3 ×3 ×3 ×32 + BN + MaxPool 2 ×2 (stride =2))	2 (3 ×3 ×3 ×64 + BN + MaxPool 2 ×2 (stride =2)), 2 (3 ×3 ×3 ×128 + BN + MaxPool 2 ×2 (stride=2)), Flatten	3 FC(512, 128, 2)	–	3	–	Dropout (0.5), L2	BN; HM; Z-norm	After CL; At input; At input	CE	Adam	50	–	–	–
128	Cantürk (2021a)	Classif.	–	AlexNet, GoogleNet (pre-trained)	AlexNet (pretrained)	SVM, KNN	–	–	–	–	–	–	–	–	–	–	–	–
129	Bhan et al. (2021)	Classif.	–	LeNet + BN	–	–	ReLU	–	–	–	BN	After CL	CE	Adam	30	–	–	–
130	Balaji et al. (2021)	Classif.	10	1 (LSTM (50) + Dropout (0.2))	3 (LSTM (50) + Dropout (0.2))	3 FC (50, 25, 4)	Tanh, Softmax	3	–	L2	–	–	CE	Adam	50	0.001	0.9	–
131	Loh, Ooi, et al. (2021)	Classif.		1 (5 ×5x16 (constraint = 3) + Dropout (0.2))	1 (3 ×3 ×32 (constraint = 3) + MaxPool), Flatten	2 FC(512, 3)	ReLU, Softmax, Sigmoid	2	–	Const-raint	–	–	–	Adam	–	0.001	0.01	–

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
132	Yadav (2021)	Classif.	–	Dropout (0.05)	1 (5 ×5x5 ×4 + Dropout (0.10) + MaxPool 4 ×4 ×4), 2 (4 ×4 ×4 ×8 + Dropout (0.15) + MaxPool + Dropout (0.20)), Flatten	2 FC (16, 16, 2)	Softmax	3	Dropout (–)	Dropout (0.20)	Smoothing, Temporal filtering	At input	–	–	–	–	–	–
133	Varrecchia et al. (2021)	Classif.	1 - 5	1 FC (9)	1 - 3 FC (20-50)	1 FC (2)	Sigmoid	1 - 5	–	–	–	–	MSE	LM	1000	–	–	GS
134	Shaban (2021)	Classif.	12	2 FC(512)	9 FC (128, 128, 64, 64, 32, 32, 16, 16, 8)	1 FC(1)	Sigmoid	–	–	–	–	–	–	CGD	335, 306, 213	–	–	GS
135	Ashfaque et al. (2021)	Mostafa Classif.	13	1 (4 ×4 ×8 + MaxPool 2 ×2 + Dropout (0.2))	1 (4 ×4 ×16 + MaxPool 2 ×2 + Dropout (0.2)), 1 (4 ×4 ×32 + MaxPool 2 ×2 + Dropout (0.2), 1 (4 ×4 ×64 + MaxPool 2 ×2 + Dropout (0.2)), Flatten	3 FC (100, 50, 3)	LRelu (0.3), Softmax	3	–	Dropout (0.2)	0-1	–	–	RMSProp	–	0.0001	–	–
			5	1 BiLSTM(–)	3 BiLSTM(–)	1 FC (3)	Tanh, Softmax	1	–	–								
			2	1 FC (10)	–	1 FC (3)	Softmax	2	–	–								

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Table 2 (*continued*).

Sr. No	Study	Problem type	# of layers	# of hidden layers (Hidden Units)			Activation	# of dense layers	Regularization		Normalization		Loss	Optimizer	Epochs	Learning rate		Hyperparameter tuning
				Initial	Middle	Last			Conv layers	Dense layers	Type	Usage				Initial	Annealing	
136	Thummikarat and Chongstitvatana (2021)	Classif.	5	1 LSTM (50)	1 LSTM (50)	3 FC (10, 10, 1)	ReLU, Sigmoid	3	–	–	–	–	CE	Adam	200	0.01	–	GS
137	Leung et al. (2021)	Classif.	–	1 CLSTM (3 ×3 ×64) + BN	1 (3 ×3 ×3 ×64 + GAP)	Concat, Dropout (0.5)	1 FC (1)	ReLU, Sigmoid	–	–	BN	After CL	MAPE, MAE, PCC	Adam	200	–	–	GS
			–	1 LSTM (64)	VGG16, ResNet50, DenseNet121, InceptionV3													
			–	1 LSTM (64)														

HWI - Handwriting images, **MSA** - Multiple System Atrophy, **LOOCV** - Leave one out Cross-validation, **PSP** - Progressive supranuclear palsy, **MSE** - Mean squared error, **CV** - Coefficient variance, **AI** - Asymmetry index, **AUC** - Area under curve, **STFT** - Short time Fourier transform, **LFP** - Local field potential, **MAE** - Mean absolute error, **DT** - Decision tree, **NB** - Naive Bayes, **RF** - Random forest, **GA** - Genetic algorithm, **AE** - Autoencoder, **Sp-AE** - Sparse autoencoder, **KCFW** - K-means clustering feature weighting method, **ROC** - Receiver operating characteristics, **Kappa** - Kappa score, **RMSE** - Root mean squared error, **OPF** - Optimum path forest, **FDG** - Fluorodeoxyglucose, **Flat** - Flatten layer, **1D conv** -1D convolutional layer, **Range [0, 1]** - Values within the range [0, 1], **MaxPool** - Maxpooling operation, **Pool** - Pooling operation, **Conv** - Convolutional layer, **LR** - Learning Rate, **Adam** - Adaptive moment estimation, **RMSprop** - Root mean square propagation, **BN** - Batch normalization, **SGDM** - Stochastic gradient descent with momentum, **arch** - Architecture, **FC layer** - Fully connected layer, **opt** - Optimization, **Classif.** - Classification, **Reg** - Regression, **Pred** - Prediction, **ReLU** - Rectified linear unit, **GD** - Gradient descent, **HL** - Hidden layer, **IPD** - Idiopathic Parkinson's disease, **CDA** - Contrastive divergence algorithm, **tanh** - Hyperbolic tangent, **NADAM** - Nesterov Adam, **GPA** - Gradient projection algorithm, **DB** - Dense block, **AE** - Autoencoder, **Norm** - Normalization, **SCG** - Scaled conjugate gradient descent, **Softmax Reg. layer** - Softmax regression layer, **KL** - KL divergence, **CAE** - Convolutional autoencoder, **SubGD** - Sub gradient descent, **LF** - Logistic function, **GD** - Gradient descent, **GAP** -Global average pooling, **RR** - Readout regularization, **SR** - Spectral radius, **SELU** - Scaled exponential linear unit, **LM** - Levenberg–Marquardt optimization, **BLSTM** - Bidirectional LSTM, **Clf** - Classifier, **Gen** - Generator network, **Disc** - Discriminator network, **SE** - squared error, **WD** - Weight decay, **HM** - Histogram matching, **Z-norm** - Z-normalization, **CL** - Convolution layers, **LRelu** - Leaky ReLU, **MAPE** - Mean Absolute Percentage Error, **PCC** - Pearson Correlation Coefficient, **KNN** - K Nearest Neighbours algorithm.

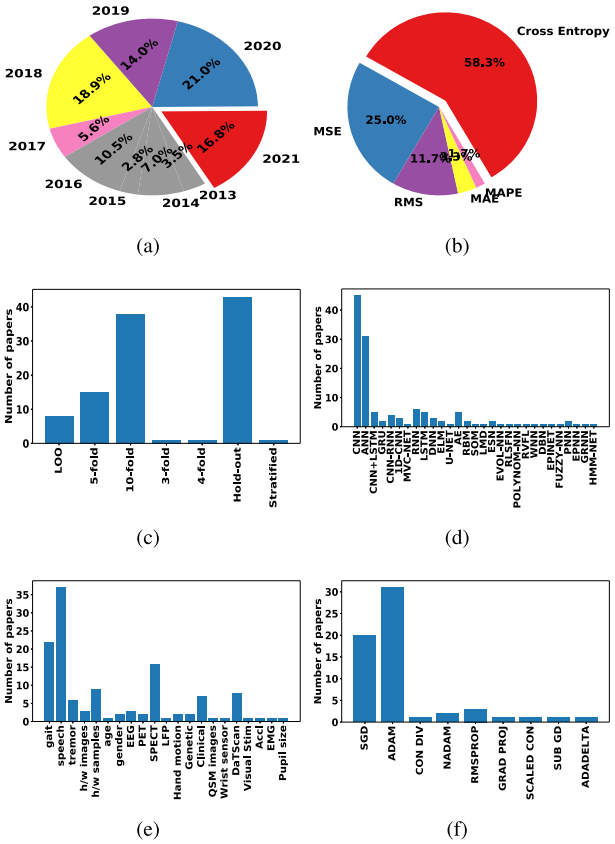


Fig. 1. (a) Year (b) Loss function (c) Cross-validation methods (d) Architecture (e) Modality, and (f) Optimizer.

The papers discussed above have a subject-wise division of training and testing data, which helps in avoiding the data leakage problem. However, a disadvantage of these models in the sheer number of model parameters, which can be significantly greater than 2D convolutional models.

1D-CNN models: Zanca et al. (2020) used batch normalization after the input layer to normalize the inputs in a data-dependent manner. A 1D 3×1 kernel follows a 1D 5×1 kernels in the subsequent layer. Um et al. (2017) utilized data augmentation techniques for time series data to increase training sample size and improve model generalization performance. Batch normalization is also used after convolution layers. The use of smaller kernels with large sample sizes allows building a more in-depth architecture. Khatamino et al. (2018) used strided max-pooling operations. Dropout is also used to reduce overfitting. El Maachi et al. (2020) utilize ensembles to improve the proposed model's generalization performance. The early stopping technique is used as a way to prevent overfitting, along with the use of dropout. Early stopping has been found helpful in preventing the network from overfitting (Caruana, Lawrence, & Giles, 2001). However, when to stop the training is still an issue that has to be dealt with carefully (Yao, Rosasco, & Caponnetto, 2007). Qin et al. (2019) proposed an architecture that leveraged L2 regularization for preventing overfitting and used a sizeable 1D convolution kernel of size 12×1 throughout the network. Bikias et al. used a larger sized 1D convolution kernel for capturing information from a larger context. A GAP layer was used before flattening layer to combine features from each channel. The usage of GAP helped to reduce the dimension of the feature map while maintaining information from individual features in each channel. In order to properly train the model, early stopping was also used.

CNN-RNN models: Sigcha et al. (2020) used time distributed convolution layers with 4×1 1D convolution kernels. The time distributed outputs were passed to the LSTM layer as a sequence along with Gradient clipping. Random search is used for hyperparameter tuning. The usage of gradient clipping aids in training the network as the LSTM by preventing the explosion of gradients (Kanai, Fujiwara, & Iwamura, 2017). The random search is a highly effective technique for training deep networks with a large hyperparameter search space (Bergstra & Bengio, 2012). Oktay and Kocer (2020) also used a relatively shallower vanilla CNN-LSTM architecture on tremor signals. Kollia et al. (2019) used a ResNet-50 pretrained CNN for feature extraction, followed by a GRU. Zhao et al. (2018) proposed a multi-path architecture to learn from images as well as time-series signals simultaneously. The outputs from the multiple paths are fed to the dense layer, which produces the final classification results. A convolution kernel of 5×5 is used for images. Whereas, for time series features, a two-layer LSTM model is followed by a softmax layer. Demir et al. (2021) converted the image feature maps into a sequence by using a sequence folding layer which was then passed on to an

Table 3
Pros and cons of 2D CNN models for PD diagnosis and classification.

Paper	Pros	Cons
Naseer et al. (2020)	Transfer learning. Usage of momentum and learning rate schedule. Usage of data augmentation.	Possible data leakage as there is no mention of subject wise dataset division.
Xue and Xin (2019)	Utilization of sparse structure within a CNN.	No utilization of transfer learning. Effect of momentum and learning rate schedule not studied.
Hwang et al. (2018)	Usage of joint feature learning. Usage of hybrid structure for input reconstruction. Usage of data augmentation.	Small dataset.
Pereira et al. (2017)	Usage of ensemble. Usage of subject-wise data partitioning.	No feature learning on raw time series data.
Afonso et al. (2019)	Usage of recurrence plots.	No transfer learning.
Sivaranjini and Sujatha (2019)	Usage of pretrained model. Usage of momentum.	Data leakage.
Zhang et al. (2018)	Usage of spectrograms. Usage of joint time–frequency analysis algorithm.	No transfer learning. No inclusion of demographic information.
Golshan et al. (2020)	Usage of 2D convolutions.	Small dataset. No transfer learning. No skip connections.
Johri and Tripathi (2019)	Usage of lightweight CNN.	No transfer learning. No skip connections.
Pang et al. (2020)	Usage of pretrained model to extract features.	Small dataset.
Wodzinski et al. (2019)	Learns better feature representation. Usage of transfer learning.	Less domain adaptation.
Xiao et al. (2019)	Usage of pretrained model for feature extraction. Usage of feature fusion.	Data leakage.
Kim, Wit, and Thurston (2018)	Usage of vanilla CNN.	Small dataset. No skip connections.
Kim, Lee, et al. (2018)	Subject wise data division. Usage of data augmentation. Usage of pretrained model.	Less domain adaptation.
Mohammed et al. (2020)	Subject wise data division. Usage of data augmentation. Usage of image normalization. Usage of dropout.	No skip connections. No transfer learning.
Yasaka et al. (2021)	Subject wise data division. Usage of data augmentation. Strong focus on clinical. interpretability.	No transfer learning No skip connections.
Cantürk (2021b)	Usage of pretrained models.	Need to test for more signal to image conversion methods.
Loh, Ooi, et al. (2021)	Usage of weight constraint. Lightweight model.	Less data.

LSTM layer. The BN layer was used after convolution for feature map normalization. For further regularizing the network, dropout layers were used in the dense layers. Input feature normalization was also performed, which further helped the model's learning capability. Sivakumar et al. extracted features using a pretrained LeNet model. If the extracted features were from the PD class then the extracted features were then passed on to the LSTM layer. Mostafa et al. also used dropout layers after convolution layers. Four

Table 4
Pros and cons of 3D CNN models for PD diagnosis and classification.

Paper	Pros	Cons
Choi et al. (2017)	Usage of strided convolutions. Strong focus on domain adaptation. Strong focus on CI.	No hyperparameter tuning. No skip connections.
Choi et al. (2020)	Usage of strided convolutions. Usage of strided pooling layers.	No skip connections. No transfer learning.
Pianpanit et al. (2020)	Usage of strided convolutions. Usage of strided pooling layers. Strong focus on model interpretability.	No skip connections. No transfer learning.
Ortiz et al. (2019)	Novel input features. Usage of transfer learning.	No skip connections.
Adams et al. (2018)	Usage of strided convolutions.	No skip connections. No transfer learning.
Esmaeilzadeh et al. (2018)	Strong usage of regularization schemes. Usage of data augmentation. Usage of LR schedule.	No skip connections. No transfer learning.
Khojasteh et al. (2018)	Usage of data augmentation. Usage of lightweight CNN.	Unable to model sequences.
Adams et al. (2021)	Usage of image normalization and label smoothing and augmentation. Usage of parallel branches.	Small dataset. No focus on interpretability.
Yadav (2021)	Usage of Bayesian network.	Small dataset. No focus on interpretability.
Vyas et al. (2021)	Lightweight network.	No use of skip connections.
Heidarivincheh et al. (2021)	Robust to missing modalities. Usage of generative network.	Small dataset. Take samples from free living/clinical tests. Need to include more sensors.

different feature views were used to train four independent CNNs with similar architecture. Two BiLSTM based networks were also trained on different feature views. The final classification was performed by an ensemble of all the different models.

RNN based models: Ashour et al. (2020) and Masiala et al. (2019) used an LSTM architecture for modelling time-series signals. The latter used two layers deep LSTM network followed by dense layers. Whereas, the former consisted of one LSTM layer followed by a dense layer. Lin et al. (2020) used dropout layers after LSTM encoder layer. The encoder layer after training is used as input to the dense layers for classification. Shahtalebi et al. (2020) and Ribeiro et al. (2019) used a GRU based network. Szumilas et al. (2020), Tang et al. (2019), Lacy et al. (2018) and Pereira et al. (2017) use other variants of shallow recurrent neural networks. Balaji et al. extracted features from a VGRF signals. The features were passed on to four consecutive LSTM layers, each with 50 units. The feature map from the final LSTM unit was passed through a dropout layer, which produced a regularization effect. Further regularization was performed using L2-regularization. Thummikarat and Chongstitvatana (Thummikarat & Chongstitvatana, 2021) used two LSTM layers with 50 units on handcrafted features

ANN based models: Khan et al. (2018) proposed an evolutionary neural network ensemble model. and used the Gasen-b ensemble pruning method to prune the ensemble, which resulted in better generalization as well as the reduction in computational costs. Gürüler (2017) proposed a novel feature weighting method using K-means clustering for the complex-valued ANN. Akyol (2020) used six dense layers deep network. The authors of works (Grover et al., 2018; Vital et al., 2020), have focused on ANN for classification, mainly consisting of one hidden and one output layer. Babu and Suresh (2013) used microarray gene expression data and independent component analysis (ICA) with 'projection based learning for meta-cognitive radial basis function network (PBL-McRBFN)'. Hosseini and Makki (2013) have used PCA with auto-associative NN for the PD detection and diagnosis. Hariharan et al. (2014) introduced detection of PD using PNN, and GRNN with speech data. Similarly, Peker et al. (2015) used CVANN for diagnosing PD with the integration of the extracted features using mRMR features from the speech signal.

MLP based models: Sharma and Giri (2014) used MLP based ANN to discriminate PD subjects. Beheshti et al. presented an improved version of ANN such as CAPSO-MLP, PSO-MLP, GSA-MLP, ICA-MLP to recognize the different kind of disease, including PD. Huang et al. (2020) used MLP and its ensemble learning models for the classification of PD using PCA, IStat, and multilinear PCA features extractor over SPECT imaging.

FB-ANN based models: Islam et al. (2014) used FB-ANN for early detection and diagnosis of PD over voice disorder speech dataset. Salmanpour et al. (2021) have used several algorithms and RBF to classify the PD, utilizing radiomics based imaging information.

Table 5
Pros and cons of 1D CNN models for PD diagnosis and classification.

Paper	Pros	Cons
Zanca et al. (2020)	Usage of data dependent input normalization.	Unable to learn long-term dependencies.
Um et al. (2017)	Usage of data augmentation. Usage of smaller sized kernels.	Inability to deal with noisy labels. No skip connections.
Khatamino et al. (2018)	Usage of strong regularization scheme.	Unable to learn long-term dependencies in time series signals.
El Maachi et al. (2020)	Usage of strong regularization scheme. Usage of ensembles.	No skip connection within parallel 1D convolutional branches.
Qin et al. (2019)	Usage of L2 regularization Usage of a lightweight CNN. Usage of similarity learning to boost generalization performance.	Usage of handcrafted features. No skip connections.
Oh et al. (2018)	Usage of a lightweight CNN. Usage of dropout regularization.	No skip connections.
Martinez-Murcia et al. (2017)	Usage of a lightweight CNN.	Unable to model sequential data.
Zhang et al. (2020)	Usage of data augmentation. Usage of batch Normalization. Strong focus on CI.	No usage of transfer learning. No skip connections.
Bikias et al. (2021)	Features learnt from data. Lightweight network.	Small dataset Less focus on CI.

Table 6
Pros and cons of CNN-RNN models for PD diagnosis and classification.

Paper	Pros	Cons
Li et al. (2020)	Usage of SE blocks. Usage of attention mechanism. Usage of input segmentation and augmentation.	No transfer learning. No skip connections.
Wingate et al. (2020)	Usage of pretrained CNN for feature extraction. Usage of bidirectional layers. Strong focus on domain adaptation.	No inclusion of other clinical and historical data.
Sigcha et al. (2020)	Usage of time distributed convolutions. Usage of gradient clipping. Usage of random search for hyperparameter tuning.	No data augmentation. No transfer learning.
Oktay and Kocer (2020)	Usage of a lightweight architecture.	Small dataset size. No data augmentation. No transfer learning.
Kollia et al. (2019)	Usage of pretrained CNN for feature extraction. Usage of bidirectional layers. Strong focus on domain adaptation.	No inclusion of other clinical and historical data.
Zhao et al. (2018)	Usage of a multi-path architecture to boost generalization performance. Usage of feature fusion.	No usage of more advanced hyperparameter tuning techniques like random search, Bayesian optimization.
Demir et al. (2021)	Usage of end-to-end image to sequence conversion and learning. Usage of BN.	Relatively more computationally intensive.
Ashfaque Mostafa et al. (2021)	Usage of ensemble. Robust to noisy data.	Computationally intensive.

Table 7
Pros and cons of RNN models for PD diagnosis and classification.

Paper	Pros	Cons
Quan et al. (2021)	Use of grid search. Use of bidirectional layers.	Handcrafted features
Ashour et al. (2020)	More computationally efficient method for modelling gait signals.	No usage of bidirectional scheme Very shallow network.
Masiala et al. (2019)	Usage of a deeper network.	Unable to learn complex relationship dependencies Relatively simpler feature learning method.
Lin et al. (2020)	Autoencoder–decoder architecture. Usage of dropout.	Handcrafted features. Small dataset size.
Shahtalebi et al. (2020)	Usage of bidirectional layers for online and offline learning of sequences. Can model long-term sequences. Usage of grid search for hyperparameter tuning.	Small dataset size. No inclusion of other modalities.
Ribeiro et al. (2019)	Can model long-term dependencies. Usage of grid search for hyperparameter tuning. Usage of early stopping.	Small dataset size. Handcrafted sampling rates and durations.
Szumilas et al. (2020)	Usage of ensembles. Usage of LASSO regularization scheme. Model consumes lesser memory with faster training,	Small dataset size. Very shallow network.
Tang et al. (2019)	Usage of lightweight shallow network to give on-device predictions.	Not very effective in power usage and computational cycle time for delivering real time predictions.
Lacy et al. (2018)	Model consumes lesser memory with faster training.	Hyperparameter tuning of ESNs is not easy.
Pereira et al. (2018)	Can learn more robust feature representations Usage of grid search.	Handcrafted features. Relatively shallow network
Gallicchio et al. (2018)	Faster training. Less computationally expensive. Usage of grid search for hyperparameter tuning. Usage of ensembles.	No inclusion of signals from multiple modalities.
Gallicchio et al. (2018)	Ability to model long sequences.	Less focus on optimal window size. No end-to-end feature learning.

Miscellaneous models: Conventional convolution operation are suited for extracting information from Euclidean spaces. In order to extract features information from the intrinsic manifold structure present in MRI scans, Bouza et al. (2020) proposed manifold valued convolutions (MVC-Net) and applied it for diagnosing PD.

Zhang (2017) used SAE for dimensionality reduction. Peng et al. (2019) also utilized an AE for reducing the dimension of features input from a node2vec model. The usage of sparsity constraint in AE may lead to better performance as values of the irrelevant features are brought close to zero, thereby, reducing their effect on the model. Shi et al. (2018) cascaded multiple RVFL layers to create a five-layer deep network and trained the network in the presence of privileged information (PI) that is not available during testing. The PI is generated from a single modality and does not require the presence of a different modality, which makes it beneficial for cases where multiple modalities are not available. The cascaded structure helps the model to learn more refined features. Passos et al. (2018) used the ResNet50 (He et al., 2016) architecture for feature extraction. To reduce the dimension, principal component analysis (PCA) was used, and further classification was performed with the help of other classifiers like RF and SVM.

Further, detailed discussions about pros and cons of some models covered in this study is given in the Tables 3–9.

Table 8
Pros and cons of ANN models for PD diagnosis and classification.

Paper	Pros	Cons
Gharehchopogh and Mohammadi (2013)	Usage of RBF function. Usage of LM. Validated the model in two-way.	Limited data. Imbalanced dataset. Model achieved less accuracy.
Hossen (2013)	Usage of wavelet-decomposition. Usage of soft-decision algorithm. Usage of MLP. Validated the results in five-way.	Limited data. Model achieved less accuracy.
Salmanpour et al. (2021)	Usage of Radiomics features. Usage of data-characterization algorithms. Multi-class classification. Usage of DRA.	Imbalanced dataset. Not reported the results in *Acc, Sen, Spec, F1-score.*

4.2. Further insights from the study conducted

The Fig. 1 shows some insights extracted from the literature survey. We can observe the emergence of neural networks for PD diagnosis over the years through Fig. 1(a). Fig. 1(c) shows the usage of loss functions in the researches surveyed in this paper. The cross-entropy (CE) loss is the most widely used loss function. This is because most of the papers focus on a classification problem. Out of the papers focusing on regression, the most widely used loss function is the root mean squared (RMS) loss function.

The researchers have mostly used conventional ANN and CNN, as shown in Fig. 1(d). This is because most researchers have used CNNs to generate high-level abstract features automatically learned from the data itself. Many layers of convolution operation lead to better features for images as well as time-series data. The most widely used modality for PD diagnosis is speech and gait 1(e). As vocal impairments and movement impairments primarily accompany PD, most researchers focus on deriving features from these modalities.

5. Future directions

We can observe from the Table 2 that only a few researchers use strong regularization and normalization techniques in deep networks. Regularization and normalization techniques have proven highly effective in deep learning literature. Thus, more efforts need to be given on incorporating state-of-the-art regularization and normalization techniques to improve model generalization. We can also observe that only a few deep models utilize the power of skip connections. However, skip connections have lately become a highly effective strategy to improve model depth and performance. Thus, more efforts should also be given to developing deep models that leverage skip connections.

We also observe from the Table 2 that only a few papers consider performing learning rate annealing and gradient clipping, which have shown potential in boosting model performance. Therefore, incorporating these techniques should also be regarded as notable in future works and should also be given more advanced learning paradigms like multitask learning and transfer learning more importance in the future.

We can also observe from the Table 2 that not much effort is given to capture more diverse information from different feature views to boost model performance. Hence, it should also focus on multimodal and multi-view features and increase diversity in the model using ensemble strategies.

Hence, more advanced deep learning ensemble strategies should also be given focus to improved generalization performance of the model. More efforts should also be given in developing models with multiple paths and heads to leverage the advantages of auxiliary learning. In order to provide unbiased information to the model, care should also be taken to avoid data leakage, especially in studies from which multiple feature views are extracted from a single subject. The training and testing datasets should be divided subject-wise in order to prevent data leakage.

Most of the models studied in this paper do not focus on model interpretability. However, interpreting the outcomes of deep models is of prime importance for clinical usage of the models. Hence, it is of utmost importance that more efforts are given in interpreting the models.

From the Table 1, we can observe that only a few researchers (Choi et al., 2017; Wu et al., 2018) focus on the problem of SWEDD classification. SWEDD subjects are clinically diagnosed as PD but do not show a dopaminergic deficiency in brain scans. This causes difficulty in diagnosis of scans without evidence for dopaminergic deficit (SWEDD) subjects. The similar nature of brain scans of SWEDD subjects and normal subjects causes considerable difficulty in classification. This problem should be focused more on the future. SWEDD belongs to a category of an atypical subgroup of PD. Similarly, IPD and other atypical PD syndromes need to focus on automated diagnosis in the future, as they pose a tougher diagnosis challenge.

The more accurate detection of FoG is also an important aspect that should be focused upon in future works. More rigorous evaluation of the proposed methods for detection of FoG needs to be done with a large amount of data, in order to account for

Table 9
Pros and cons of miscellaneous models for PD diagnosis and classification.

Paper	Pros	Cons
Akyol (2020)	Usage of Growing and pruning approach for finding relevant architecture.	No convolution layers.
Bouza et al. (2020)	Usage of geometric deep learning. Strong theoretical description.	Relatively shallow structure
Zhang (2017)	Better feature representation. Fast classification through KNN classifier.	Unable to deal with noisy speech samples. Unable to deal with evolving data.
Shi et al. (2018)	Utilization of privileged information. Utilization of cascaded architecture.	Handcrafted features.
Passos et al. (2018)	Usage of pretrained network for feature extraction. Usage of hybrid architecture.	No domain dependent feature representation
Vásquez-Correa et al. (2018)	Usage of both 2D and 1D lightweight CNN for learning features from multiple modalities. Usage of feature fusion.	Unable to model long and short term sequences.
Grover et al. (2018)	Usage of lightweight vanilla feed forward network.	Handcrafted features. No convolutional layers
Nilashi et al. (2018)	Usage of SOM for clustering.	Hand crafted features.
Peng et al. (2019)	Features extracted from gene expressions. Usage of autoencoder structure for feature representation learning.	No convolutional layers. Relatively shallow network architecture.
Kadam and Jadhav (2019)	Usage of multi-path architecture for creating feature ensembles.	Hand crafted features No convolutional layers.
Gong et al. (2018)	Usage of margin maximization with margin variance minimization.	Handcrafted features. Relatively shallow network. No convolutional layers.
Wroge et al. (2018)	Usage of vanilla feed forward neural network.	No convolutional layers. Handcrafted features.
Wang et al. (2020)	Usage of vanilla feed forward neural network.	No convolutional layers. Handcrafted features.
Kaur et al. (2020b)	Usage of vanilla feed forward neural network. Strong focus on network topology optimization.	No convolutional layers. Handcrafted features.

different patterns of FoG. The problem can also be approached from the perspective of class imbalance, which can further enhance the performance and lead to various novel time–frequency analysis techniques for deep neural networks.

The reader can also view the information contained in supporting tables to seek new future directions with respect to deep learning models. The points mentioned in the pros column can contribute to good practices that the reader can include in his research. While, the points mentioned in the cons column can become a point for the researcher to contribute on and can become the direction on which he could work.

The reader could also correlate the information of supporting tables with Tables 1 and Table 2 for an in-depth analysis. We provide three examples of how this can be done:

- Example 1: From Table 2, we can observe that Mohammed et al. (2020) develop an AlexNet like model which uses a kernel size of 3×3 all throughout. The smaller kernel size enables the model to reduce the computational and space complexity. From supporting Table 3, we observe that Mohammed et al. (2020) also performed data augmentation and image normalization prior to feeding the input to the model. We also observe from the 'cons' column that the model does not use transfer learning. Transfer learning has found to be a highly effective way of getting better results in deep learning. Thus, one possible direction to improve upon this work would be to develop a robust transfer learning method. Another point from 'cons' column we can

find is that Mohammed et al. (2020) do not use skip connections within their network. However, skip connections have also been found to produce highly effective classification results as well as make the model much deeper. This could be another direction on which the model can be further developed by researchers.

- Example 2: From Table 2, we can see that Choi et al. (2017) developed a 3D CNN with $5 \times 5 \times 5$ kernels all throughout. From Table 3, we can see that their strong focus was on domain adaptation and model interpretability. However, their network lacks skip connections and because of which its depth is limited. Thus, one possible direction for improving upon this work would be to add skip connections and make the model deeper. Another line of work would be to use techniques like Bayesian optimization and hyperparameter optimization techniques to optimize the network structure on the fly, rather than developing network structure through hit and trial.
- Example 3: From Table 1, we can see that Li et al. (2020) use deep learning model on gait signals. The network structure can be seen in Table 2 and consists of squeeze-and-excitation blocks and a final attention layer. From Table 8 pros column we can observe that the model focuses strongly on data augmentation also. However, the model does not use any transfer learning technique. Thus, one possible direction could be to utilize transfer learning techniques to time series like signals that can enable one to transfer knowledge on gait signals. Another line of work could be to add further skip connections on the network.

6. Conclusion

In this paper, we performed an in-depth survey of 143 research papers that focused on diagnosing PD and its subtypes using various shallow and deep neural network architectures. We also performed an in-depth review and analysis of the architectures used by the researchers. To the best of our knowledge, such a study has not yet been performed for the computer-aided diagnosis (CAD) of PD. The comparison of various models for the CAD of PD is not easy as most of the studies have used their own dataset which is not similar to other that used in other dataset. In general, utilization of CNNs with some sort of imaging data (HWI , Spectrogram , MRI scans etc.) has led to more accurate , Gait signals etc.), usage of RNNs have been beneficial. The combination of CNN-RNN structures have also given accurate results in PD diagnosis using time-series data. Whenever clinical features (CF) is being used, the usage of shallow ANN has given better performance than other conventional ML techniques. Although the usage of CNNs has increased in recent years, the use of model interpretability techniques has not been that widespread in this field. This should be looked upon in the future as for the clinical usage of these deep models. It is highly necessary for the researchers to give the interpretation of the model to the clinicians. The focus also has to be on the early diagnosis of PD, compared to a simple PD classification from healthy subjects.

CRediT authorship contribution statement

M. Tanveer: Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **A.H. Rashid:** Data curation, Writing – original draft, Visualization, Writing – review & editing. **Rahul Kumar:** Data curation, Writing – original draft, Visualization, Writing – review & editing. **R. Balasubramanian:** Resources, Reviewing, Supervision.

Acknowledgements

This work is supported by Department of Science and Technology, INDIA under Ramanujan fellowship scheme grant no. SB/S2/RJN-001/2016, under Interdisciplinary Cyber Physical Systems (ICPS) Scheme grant no. DST/ICPS/CPS-Individual/2018/276 and Council of Scientific & Industrial Research (CSIR), New Delhi, INDIA under Extra Mural Research (EMR) scheme grant no. 22(0751)/17/EMR-II. This work was done during A. H. Rashid's tenure as a junior research fellow in the Indian Institute of Technology Indore. We gratefully acknowledge the Indian Institute of Technology Indore for providing facilities and support.

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